

SECTION 4

EVALUATION OF THE PATIENT WITH KIDNEY DISEASE

Laboratory Assessment of Kidney Disease: Glomerular Filtration Rate, Proteinuria, and Urinalysis

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CHAPTER OUTLINE

GLOMERULAR FILTRATION RATE, 679
PROTEINURIA, 688
URINE DIPSTICK, 692

URINE MICROSCOPY, 694
CLINICAL APPLICATIONS, 698

KEY POINTS

- Kidney disease is common in adults and children. Assessment of GFR, proteinuria, and urinalysis are required for detection, evaluation, and management of acute and chronic kidney disease.
- GFR is a measure of excretory function of the kidney and is the single best measure of overall kidney function in health and disease. Normal GFR is >90 mL/min/1.73 m² in young adults and children older than 2 years.
- Decreased GFR is caused by a reduction in the number of nephrons or the average single-nephron GFR.
- A GFR <60 mL/min/1.73 m² is a diagnostic criterion for acute and chronic disease, irrespective of age, and the level of GFR is used to stage the severity of acute and chronic kidney disease.
- True GFR cannot be directly measured. The reference test for measured GFR (mGFR) is urinary or plasma clearance of an exogenous filtration marker. A variety of clearance methods and markers are available.
- The initial test for GFR evaluation, estimated GFR (eGFR) from serum creatinine (eGFR_{cr}), is not sufficiently accurate for all clinical decisions. eGFR from creatinine and cystatin C (eGFR_{cr-cys}) is generally more accurate than either eGFR_{cr} or eGFR from cystatin C (eGFR_{cys}).
- Validated eGFR equations are recommended for use in practice. The most commonly used recommended eGFR equations are the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equations in adults and the Chronic Kidney Disease in Children (CKiD) equations in children.
- Most GFR estimating equations have been derived from populations with stable GFR and are less accurate if GFR is not stable.
- Albumin is the principal component of urine protein in most kidney diseases in adults. The normal value for albumin excretion rate (AER) in adults is <30 mg/day, roughly equivalent to an albumin-creatinine ratio (ACR) of approximately 30 mg/g.
- Increased albuminuria is caused by altered permeability of the glomerular capillary wall, is a marker of kidney damage, and may occur before the decline in GFR.
- Increased albuminuria is a diagnostic criterion for acute and chronic kidney disease and is used to stage the severity of chronic kidney disease.

- The reference test for albuminuria is the AER in a timed urine sample.
- The initial test for albuminuria is the ACR in an untimed ("spot") urine sample; a first morning sample is preferred. Equations are available to convert other measures of proteinuria to ACR, but their accuracy is limited at low values.
- The urinalysis includes urine dipstick and urine microscopy.
- Abnormalities in the urinalysis are markers of kidney damage, urinary tract disorders, or systemic disorders and may provide important clues to the cause of kidney disease.

Kidney disease is common in adults and children but generally causes few symptoms until the stage of kidney failure. Laboratory assessment of kidney disease is necessary for detection, evaluation, and management of kidney disease. This chapter describes key aspects of laboratory assessment, specifically glomerular filtration rate (GFR), proteinuria, and urinalysis that are part of routine clinical care. These measures are related to functional or structural alterations of the kidneys and have many clinical applications in acute kidney disease (AKD), including acute kidney injury (AKI), and chronic kidney disease (CKD) (Table 23.1). GFR is the single best measure of overall kidney function in health and disease, and evaluation of GFR is critical for many decisions in all fields of medicine. Large studies consistently demonstrate the strong relationship of GFR to clinical outcomes, and the Kidney Disease Improving Global Outcomes (KDIGO) international guidelines use GFR as one of the measures for the diagnosis and staging of AKD and CKD.¹⁻³ Proteinuria and urinalysis may indicate structural abnormalities of the kidneys and thus serve as markers of kidney damage. Albuminuria, the principal component of proteinuria in most kidney diseases in adults, is a marker of kidney damage and may occur before the decline in GFR. Albuminuria is also strongly related to clinical outcomes and is another measure for diagnosis of AKD and CKD and staging of AKD and staging of CKD. Urinalysis includes the urine dipstick and microscopy; abnormalities are markers of kidney damage, urinary tract disorders, or systemic conditions and may provide important clues to the cause of kidney disease.

GLOMERULAR FILTRATION RATE

Glomerular filtration, the ultrafiltration of plasma, is the first step in urine formation and is central to the excretory function of the kidney. Other functions of the kidney, including metabolic and endocrine functions, are necessary for volume and blood pressure regulation, acid-base and electrolyte balance, erythropoietin production, and mineral and bone homeostasis. These are discussed in other chapters. In general, reduction in GFR correlates with reduction in these other functions.

NORMAL PHYSIOLOGY

GFR is defined as the rate of filtration of plasma across the glomerular capillary wall into the Bowman space and is the product of the average of single-nephron GFR (SNGFR) times the number of nephrons. SNGFR is governed by Starling forces (net filtration pressure, ΔP) and the

properties of the glomerular capillary wall (ultrafiltration coefficient, K_f). For further discussion of the physiology of glomerular ultrafiltration, see Chapter 3.

NORMAL VALUES

The kidney filters an average of approximately 180 L of plasma per day, equivalent to 125 mL/min, with a wide range. GFR is indexed by body surface area (BSA) because kidney size is proportional to body size; this practice reduces variability in GFR among healthy individuals and allows comparisons to normative values. The normal value of GFR (125 mL/min/1.73 m²) was derived from older studies, when 1.73 m² was the average value for BSA in young adults.⁴ Studies of kidney donor candidates in the current era report a lower mean GFR, approximately 105 mL/min/1.73 m².⁵⁻⁸ The source of differences between recent and older studies is not well understood. GFR >90 mL/min/1.73 m² is considered normal for children >2 years of age and young adults, and GFR <60 mL/min/1.73 m² is one of the criteria for AKD and CKD at all ages.

Reduced GFR can be caused by decreased SNGFR due to alteration in Starling forces or decreased number of nephrons. Apart from AKD and CKD, conditions that lead to alterations in Starling forces include exercise, obesity, pregnancy, level of protein intake, hyperglycemia, surfeit or deficit of extracellular fluid, and medications that lower blood pressure, inhibit the renin-angiotensin-aldosterone system, or inhibit sodium-glucose linked transport proteins. Premature birth or intrauterine growth retardation is associated with a decreased number of nephrons and increased risks for AKD and CKD. Loss of nephrons occurs with aging,⁹⁻¹¹ but the association is variable and the cause is not fully understood (see Chapter 21).

MEASUREMENT OF GLOMERULAR FILTRATION RATE

True GFR, like many other physiologic properties, cannot be measured directly. Rather, GFR is assessed by measuring the clearance of filtration markers, substances that are eliminated primarily by glomerular filtration and whose plasma concentration varies inversely with the level of GFR (higher plasma concentration at lower GFR and lower plasma concentration at higher GFR). An ideal filtration marker for GFR measurement would have the following characteristics:

- Short equilibration time between plasma and final volume of distribution

Table 23.1 Clinical applications of laboratory assessment of kidney disease

Topic/Application	Population	GFR	Proteinuria	Dipstick	Urine microscopy
Significance	All	Overall index of kidney function	Albuminuria as a marker of kidney damage	Initial test for markers of kidney damage	Confirmatory test for markers of kidney damage
Defining kidney disease	AKI	↑Scr by 0.3 mg/dL in 48 hours or by 50% in 7 days, or oliguria >4 hours	NA	NA	NA
	AKD	AKI or GFR <60 or ↓GFR by 35% for <3 months	ACR >30 for <3 months	Present for <3 months	Present for <3 months
	CKD	GFR <60 for >3 months	ACR >30 for >3 months	Present for >3 months	Present for >3 months
Staging kidney disease (severity)	AKI	Stages 1-3 defined by changes in Scr, urine output, and initiation of KRT	NA	NA	NA
	AKD	As for AKI if AKI present As for CKD if AKI absent	NA if AKI present As for CKD if AKI absent	NA	NA
	CKD	G1-G5 defined by level of GFR	A1-A3 defined by level of albuminuria	NA	NA
Screening special populations/detecting kidney disease in people at increased risk	AKI	Yes	NA	NA	NA
	AKD	Yes	Yes	Yes	Yes
	CKD	Yes	Yes	Yes	Yes
Diagnosing the cause of kidney disease	AKI/AKD/CKD	NA	Yes	Yes	Yes
Adjusting drug dosing in kidney disease	AKI/AKD/CKD	Yes	NA	NA	NA
Assessing progression of kidney disease	AKI	Yes	NA	NA	NA
	AKD/CKD	Yes	Yes	NA	Yes

AKD, acute kidney disease; AKI, acute kidney injury; CKD, chronic kidney disease; GFR, glomerular filtration rate in mL/min/1.73 m²; GP, general population; KRT, kidney replacement therapy; NA, not applicable; Yes, applicable.

Markers of kidney damage include albuminuria, urinalysis abnormalities, imaging abnormalities, pathologic abnormalities, and electrolyte disorders due to altered tubular function. Only albuminuria, urine dipstick, and urine microscopy are included in this chapter.

- Not bound to plasma proteins
- Freely filtered at the glomerulus (molecular weight <approximately 20,000 daltons)
- Not secreted or reabsorbed at the tubules
- Eliminated wholly by the kidney
- Resistant to degradation
- Easy and inexpensive to measure

None of the currently available filtration markers meets all of these requirements. Inulin, a 5200 polymer of fructose, was identified by the pioneering work of Homer Smith, and the urinary clearance of inulin remains the reference standard for measured GFR (mGFR), but inulin is not available in many countries. Except for creatinine, filtration markers used in clearance procedures for GFR measurement are exogenous substances.

CLEARANCE METHODS

The clearance (Cl) of a solute is defined as the rate at which it is eliminated from the plasma and can be measured using urinary or plasma clearance.

Urinary clearance

Urinary clearance (Cl_[u]) of an ideal filtration marker is the most direct method for measurement of GFR and is suitable for use with exogenous or endogenous filtration markers that are excreted in the urine. Urinary clearance is defined as the rate of excretion of the marker in the urine divided by the plasma concentration of the marker and can be computed according to the following formula:

$$\text{GFR} = \text{Cl}_{(u)} = \frac{\text{Urine concentration of marker} \times \text{urine flow rate}}{\text{Average plasma concentration of marker}}$$

The classic method consists of a continuous intravenous infusion of inulin to attain a constant plasma level, multiple clearance periods requiring repetitive blood and urine collections over 3 hours, oral water loading to stimulate diuresis, and bladder catheterization to ensure complete bladder emptying. In modern day practice, the procedure has been simplified by the following: administration via either intravenous or subcutaneous bolus followed by an equilibration period of 45 to 60 minutes; spontaneous

bladder emptying (preferably with bladder ultrasound to assess completeness of bladder emptying); and two to four timed urine collections, each of approximately 20 to 30 minutes, with blood samples at the time of urine collection (with averaging across the clearance periods). Intravenous administration of the marker is followed by a rapid decline in plasma concentration after the equilibration period, whereas subcutaneous administration of the marker allows for its slow release into the circulation, providing a slower decline in plasma concentration following the equilibration period. Advantages of the urinary clearance procedure are the simplicity of clearance calculations and lack of being affected by extrarenal elimination of the filtration marker. The disadvantage of the urinary clearance method is error in measuring the urine flow rate due to mistakes in timing of collections and incomplete bladder emptying.

For endogenous creatinine clearance (Cl_{cr}), the procedure can be further simplified by using a single timed urine collection (generally 24 hours) and a single measurement of plasma creatinine in the steady state or plasma measurements at the beginning and end of the urine collection in the nonsteady state. As discussed later, measured Cl_{cr} exceeds measured GFR due to tubular secretion of creatinine. A limitation of the 24-hour Cl_{cr} is its reliance on unsupervised timed urine collections, which are often inaccurate. In addition, the practical difficulties of collecting and storing a 24-hour urine collection for patients and the challenges of handling large volumes of urine for laboratories have resulted in this method being used only infrequently.

Plasma clearance

GFR can be calculated from plasma clearance ($Cl[p]$) of an exogenous filtration marker after a bolus intravenous injection. Plasma clearance is defined as the rate of elimination of the marker from plasma, divided by the rate of decline in the plasma concentration of the marker, and can be computed according to the following formula:

$$GFR = Cl(p) = \frac{\text{Quantity of the marker administered}}{\text{Area under the curve of the plasma concentration of marker over time}}$$

After an intravenous infusion of a known quantity of the marker, multiple blood samples are obtained to determine the rate of decline in plasma concentration. The early decline is due to the equilibration from plasma into the interstitial fluid (fast component). The late decline is due to excretion into the urine (slow component). The average plasma concentration is estimated using a two-compartment model that requires blood sampling early (usually two or three time points until 60 minutes) and late (one to three time points from 120 minutes onward). Most commonly this procedure is simplified using only two or more late samples to measure the slow component and the Brochner-Mortensen equation or other equations to estimate the fast component.^{12,13} Methods using a single sample collection are also available but require an additional assumption of the volume of distribution of the marker based on body size.¹⁴

Advantages of plasma clearance compared with urinary clearance include the lack of requirement for urinary

collection, which is particularly important in populations wherein bladder emptying may be impaired, such as the elderly or children with urinary tract abnormalities. There are three main disadvantages. First, a relatively long time (~5 hours) is required to determine the disappearance curve, with an even longer time required in people with very low GFR (8 to 24 hours). Shorter time periods may lead to overestimation of GFR throughout the GFR range. This limitation is more important when single sample collections are used. Second, expansion of the extracellular fluid compartment in edematous conditions causes a prolongation of the first compartment of the two-compartment curve and an overestimation of GFR. Third, extrarenal elimination of the filtration marker leads to an overestimate of GFR, which would be more apparent at lower GFR.

Imaging

GFR can also be measured by assessing the appearance of an exogenous filtration marker over the kidneys and bladder areas (dynamic renal imaging) using nuclear medicine for radiolabeled markers or magnetic resonance imaging.^{15,16} However, these methods are thought to be generally less accurate than urinary or plasma clearance methods.¹⁷ The main value of imaging-based GFR measures is to determine the relative function of each of the two kidneys or in individuals already undergoing imaging procedures.

Novel methods

Novel methods are available for transcutaneous assessment of plasma concentrations of radiolabeled or fluorescent markers to avoid blood sampling.¹⁸⁻²¹ Use of two markers, one for assessment of extracellular fluid volume and the other for assessment of urinary excretion, can shorten the duration of the procedure. These methods are not yet available in clinical practice.

EXOGENOUS FILTRATION MARKERS

Inulin is impractical for routine clinical purposes. Inulin, or the specialized assay required for its measurement, is not available in many countries. Alternative markers include iothalamate, iothexol, ethylenediaminetetraacetic acid (EDTA), and diethylenetriaminepentaacetic acid (DTPA); all can be used in urinary or plasma clearance procedures.

Iothalamate may be labeled with iodine I 125 (¹²⁵I) or used unlabeled and measured using high-performance liquid chromatography (HPLC). It is secreted by the tubules leading to overestimation of GFR. Iothexol is usually measured using HPLC. It undergoes extrarenal elimination by the hepatobiliary tract, so plasma clearance overestimates GFR.^{22,23} EDTA is usually labeled with chromium Cr 51 (⁵¹Cr).²⁴ It may be reabsorbed by the tubules, leading to underestimation of GFR. It is not available in the United States. DTPA is labeled with technetium Tc 99m (^{99m}Tc), which can dissociate from DTPA and bind to plasma proteins, resulting in underestimation of GFR. ^{99m}Tc has a short-radioisotope half-life, so external quality control is not possible.

A systematic review of comparing the accuracy of these markers to urinary clearance of inulin concluded that the strength of evidence for sufficient accuracy was strong for urinary clearance of iothalamate, moderately strong for urinary and plasma clearance of ⁵¹Cr-EDTA, and plasma

clearance of iothexol, and limited for urinary clearance of DTPA, urinary clearance of iothexol, and plasma clearance of inulin. The strength of evidence for *inaccuracy* was strong for endogenous creatinine clearance and weak for plasma clearance of DTPA.²⁵ A subsequent study recommended against plasma ^{99m}Tc-DTPA, especially when clearances were performed over 2 to 4 hours.²⁶ The most commonly used marker-clearance procedure combinations include urinary clearance of iothalamate, EDTA and DTPA, and plasma clearance of iothexol and EDTA.

ESTIMATION OF GFR

Clearance measurements are not routinely performed in clinical practice; instead, the level of GFR is usually estimated from the plasma concentration of an endogenous filtration marker. Estimated GFR (eGFR) is in widespread use in clinical practice but differs from true GFR because of the influence of physiological processes other than GFR that affect the plasma concentration of the endogenous filtration marker. These processes include generation, renal tubular reabsorption and secretion, and extrarenal elimination of the marker, and collectively, they are termed non-GFR determinants of the filtration marker. The relationships of the non-GFR determinants to GFR and plasma concentration of the filtration marker in the steady state (when GFR and non-GFR determinants are constant) are shown in Figure 23.1. GFR is inversely related to the plasma concentration of the filtration marker and modified by each of the non-GFR determinants. The inverse relation between GFR and the plasma concentration of the filtration marker allows maintenance of a constant filtered load (the product of GFR times the plasma concentration) and enables continued excretion of the marker despite reduction in GFR, but with

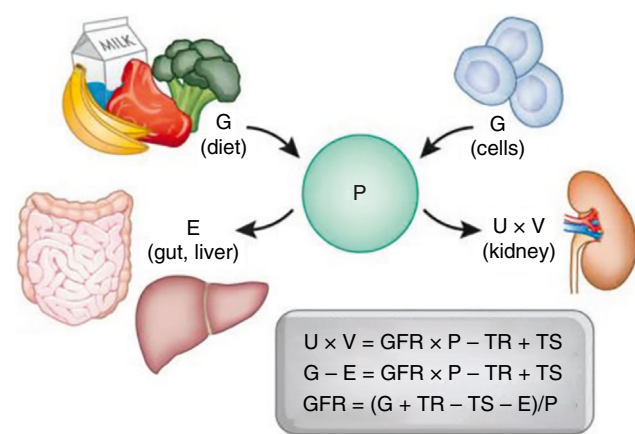


Fig. 23.1 Determinants of the serum level of endogenous filtration markers. The plasma level (*P*) of an endogenous filtration marker is determined by its generation (*G*) from cells and diet, extrarenal elimination (*E*) by gut and liver, and urinary excretion (*UV*) by the kidney. Urinary excretion is the sum of filtered load (glomerular filtration rate [*GFR*] × *P*) minus reabsorption (*TR*) plus tubular secretion (*TS*). In the steady state, generation minus extrarenal elimination equals urinary excretion. By substitution and rearrangement, GFR can be expressed as the ratio of the non-GFR determinants (*G*, *TS*, *TR*, and *E*) to the plasma level. (Permission granted from *J Am Soc Nephrol* for Stevens LA, Levey AS. Measured GFR as a confirmatory test for estimated GFR. *J Am Soc Nephrol*. 2009;20(11):2305–2313.)

the consequence of a higher plasma concentration of the marker (Fig. 23.2).

The non-GFR determinants of filtration markers are difficult to measure, so estimating equations use easily measured demographic (age, sex) and clinical variables (weight) as surrogates.²⁷ The inclusion of these variables

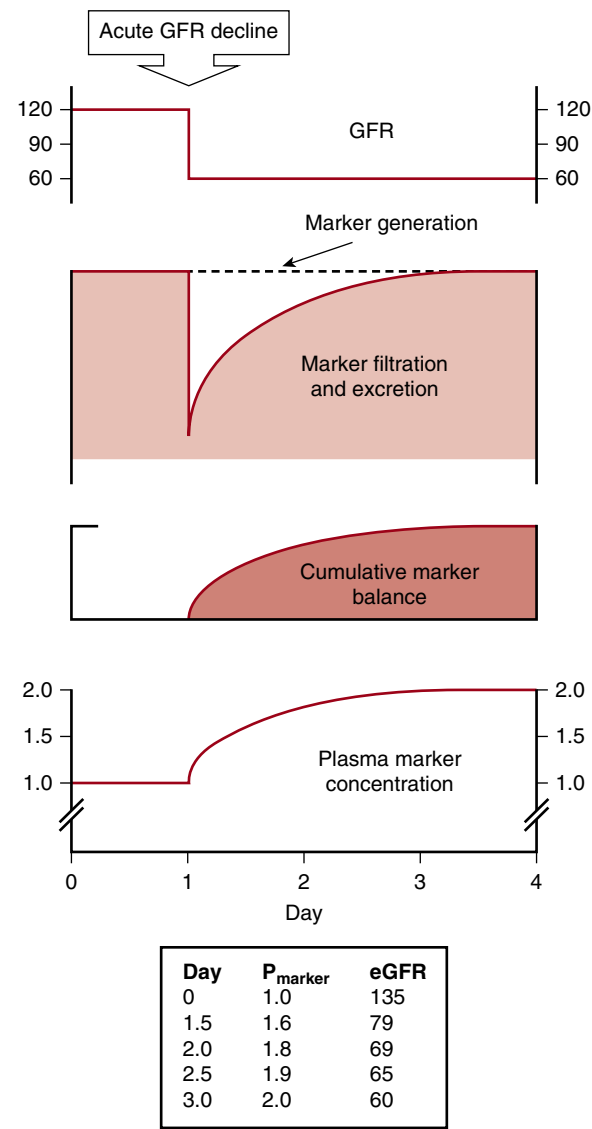


Fig. 23.2 Effect of an acute GFR decline on generation, filtration, excretion, balance, plasma marker concentration of an endogenous filtration marker (*P*_{marker}), and estimated GFR (eGFR) based on the marker. After an acute GFR decline, generation of the marker is unchanged, but filtration and excretion are reduced, resulting in retention of the marker (a rising positive balance), a rising plasma concentration, and declining eGFR (non-steady state). During this time, eGFR is higher than GFR. While GFR remains reduced, the rise in plasma concentration leads to an increase in filtered load (the product of GFR times the plasma level) until filtration equals generation. At that time, cumulative balance and the plasma level plateau at a new steady state. In the new steady state, eGFR approximates mGFR. GFR is expressed in units of milliliter per minute per 1.73 m². Tubular secretion and reabsorption and extrarenal elimination are assumed to be zero. (Permission granted from *J Am Soc Nephrol* for Stevens LA, Levey AS. Measured GFR as a confirmatory test for estimated GFR. *J Am Soc Nephrol*. 2009;20(11):2305–2313.)

minimizes systematic errors in subgroups defined by these variables and minimizes systematic differences between groups.^{28,29} GFR estimating equations convert the plasma concentration of the marker to the GFR scale and, by design, are more accurate than the plasma concentration alone and facilitate clinical decision making on the basis of the level of GFR. Thus it is recommended that clinical laboratories report eGFR whenever the filtration marker is measured. The principal limitations of GFR estimating equations are that the demographic and clinical surrogates only capture the average relations among the marker and its non-GFR determinants and that the relations among the marker and its non-GFR determinants may vary across populations.

eGFR is more accurate in the steady state than in the non-steady state (when GFR and non-GFR determinants are not constant). In the non-steady state, the rate and direction of change in the concentration of the filtration marker and in level of eGFR reflect the magnitude and direction of the change in GFR, but the change in the concentration of the marker and level of eGFR derived from the marker lags behind the change in GFR. As shown in Figure 23.2, after a fall in GFR, the decline in eGFR is less than the decline in GFR and eGFR thus exceeds GFR. Conversely, after a rise in GFR, the rise in eGFR is less than the rise in GFR and eGFR is thus less than GFR. As the serum concentration approaches the new steady state, eGFR approaches GFR. The rate of rise in the marker reflects not only the severity of the reduction in GFR but also the non-GFR determinants.

Below we first describe endogenous filtration markers in current use and investigation and then describe frequently used GFR estimating equations. Of note, serum and plasma concentrations of these markers are equivalent; next, we refer to serum concentrations.

ENDOGENOUS FILTRATION MARKERS

Creatinine

Creatinine, a 113-dalton amino acid metabolite, is a product of creatine metabolism in muscle and a nitrogenous waste product without toxicity. Creatinine is distributed throughout total body water, not bound to plasma proteins, freely filtered by the kidney, secreted by the tubules, and excreted in the urine. Serum creatinine is measured routinely as part of the basic or comprehensive metabolic panel in patients with acute or chronic illness. It can be measured by colorimetric (alkaline picrate) or enzymatic assays. Since 2008, most manufacturers of autoanalyzers use assays that are traceable to an international standard for creatinine and a reference measurement procedure.

The most important of the non-GFR determinants of serum creatinine is variation in its generation according to muscle mass and/or injury, physical activity, and consumption of meat or creatinine (Table 23.2). Since on average muscle mass differs by age and sex, lower mean serum concentrations are observed for women than men and older than younger people for the same level of

Table 23.2 Examples of conditions in which eGFR using creatinine or cystatin C are less accurate

Use	eGFR _{Cr} ^a	eGFR _{Cys} ^a
Nonsteady state (AKI)	Change in eGFR lags behind the change in mGFR (eGFR overestimates mGFR when mGFR is declining and underestimates mGFR when mGFR is rising)	Change in eGFR lags behind the change in mGFR (eGFR overestimates mGFR when mGFR is declining and underestimates mGFR when mGFR is rising)
Factors affecting generation ^b	Decreased by large muscle mass, high protein diet, ingestion of cooked meat, and creatine supplements Increased by small muscle mass, limb amputation, muscle-wasting diseases	Presumed decreased in greater adiposity, smoking, hyperthyroidism, glucocorticoid excess, and chronic inflammation, as indicated by insulin resistance, high levels of C-reactive protein and tumor necrosis factor, or low levels of serum albumin Presumed increased in hypothyroidism
Factors affecting tubular reabsorption or secretion ^b	Decreased by drug-induced inhibition of secretion	NA
Factors affecting extrarenal elimination ^b	Decreased by inhibition of gut creatininase by antibiotics Increased by dialysis, large losses of extracellular fluid (drainage of pleural fluid or ascites)	Increased by large losses of extracellular fluid (drainage of pleural fluid or ascites)
Range	Less precise at higher GFR, due to higher biological variability in non-GFR determinants relative to GFR and larger measurement error in Scr and GFR	Less precise at higher GFR, due to higher biological variability in non-GFR determinants relative to GFR, and larger measurement error in Scys and GFR
Interference with assays	Spectral interferences (bilirubin, some drugs) Chemical interferences (glucose, ketones, bilirubin, some drugs)	At high levels, may be susceptible to hook effects

^aEffects eGFR, opposite effects on serum creatinine and serum cystatin C.

^bGeneration, tubular reabsorption and secretion, and extrarenal elimination are non-GFR determinants. Errors in eGFR_{Cr}-cys related to non-GFR determinants of creatinine or cystatin are hypothesized to be less than for eGFR_{Cr} and eGFR_{Cys} alone.

AKI, Acute kidney injury; eGFR_{Cr}, glomerular filtration rate estimates based on serum creatinine; eGFR_{Cys}, glomerular filtration rate estimates based on cystatin C; GFR, glomerular filtration rate; mGFR, measured glomerular filtration rate; Scr, serum creatinine; Scys, serum cystatin C.

mGFR. The observation that rhabdomyolysis is sometimes associated with a faster rise in serum creatinine than other causes of acute GFR decline has been suggested to be due to release of creatine and phosphocreatine that is converted into creatinine. Secretion of creatinine by the renal tubule occurs by the basolateral organic cation secretory pump (OCT) and the luminal solute carrier family 47 member 1 (SLC47A1) transporter (also known as multidrug and toxin extrusion 1, or MATE1). Secretion can be inhibited by other organic cations and is affected by many drugs.³⁰ Extrarenal elimination of creatinine may occur when bacterial overgrowth in the intestines leads to increased degradation of creatinine by bacterial creatininase activity, which may be inhibited by broad-spectrum antibiotics.³¹ The contribution of tubular secretion and extrarenal elimination of creatinine to renal creatinine elimination rises when the serum creatinine is higher, so drug-induced inhibition of these processes leads to a greater increase in serum creatinine when the GFR is lower.

Cystatin C

Cystatin C, a low-molecular-weight (13,343-dalton) protein, consists of 120 amino acid residues in a single polypeptide chain.³² Cystatin C regulates the activities of cysteine proteases to prevent uncontrolled proteolysis and tissue damage.^{33,34} Cystatin C is generated by all nucleated cells, and its distribution is limited to the extracellular space. It is freely filtered by the glomerulus and is not secreted; it is reabsorbed and catabolized by proximal tubule cells, so only a small amount is normally excreted. Increased urinary excretion of cystatin C is a marker of proximal tubular damage. Serum cystatin C can be measured by immunoassays, and some assays can be included on an autoanalyzer. An international standard for cystatin C is now available, but there is not yet a reference measurement procedure.³⁵

The non-GFR determinants of serum cystatin C are less well understood than the non-GFR determinants of serum creatinine, but it is appreciated that serum cystatin C concentrations are less affected than serum creatinine by sex and age but more affected than serum creatinine by adiposity, smoking, hyperthyroidism, glucocorticoid excess, and chronic inflammation, as indicated by insulin resistance, higher levels of C-reactive protein and tumor necrosis factor, or lower levels of serum albumin, independent of GFR (Table 23.2).³⁶⁻⁴⁷

Urea

Urea, a 60-dalton metabolite, is the end-product of amino acid degradation by the urea cycle. Urea is the primary nitrogenous waste product excreted in the urine and is toxic at high serum concentrations (blood urea nitrogen >150 mg/dl). Urea is distributed throughout total body water, not bound to proteins, freely filtered by the glomerulus, reabsorbed by the tubules by passive diffusion, and excreted in the urine. Serum urea is often expressed as SUN or blood urea nitrogen (BUN).

Urea is measured routinely in clinical practice but is no longer interpreted primarily as a filtration marker because of the large number of non-GFR determinants, and alterations in SUN should be interpreted accordingly. For example, high concentrations of SUN can be due to decreased GFR; increased generation due to high protein intake, gastrointestinal

bleeding due to absorption of amino acids from blood in the gastrointestinal tract, and high catabolic states such as those associated with glucocorticoid therapy; and increased reabsorption, particularly at low urine flow rates. Conversely, low concentrations of serum urea nitrogen (SUN) can be due to increased GFR; decreased generation due to decreased protein intake and chronic liver disease; and decreased reabsorption, particularly at high urine flow rates.

Other metabolites and low-molecular-weight proteins

Many metabolites and low-molecular-weight proteins have been identified as candidate filtration markers because of the strong inverse correlations of their serum concentrations with measured GFR and are undergoing investigation to determine their clinical utility. Low-molecular-weight proteins that have been extensively studied include β_2 -microglobulin (B2M), an 11.8-kDa protein produced by all nucleated cells and commonly used as a prognostic marker for B-lymphocyte and plasma cell neoplasms, and β -trace protein (BTP), a 23–29-kDa protein produced in the central nervous system and by other tissues and used to detect cerebrospinal leaks.⁴⁸ Metabolites of interest include N-acetylthreonine, pseudouridine, N-acetylserine, meso-erythritol, arabinol, myo-inositol, N-acetylalanine, 3-indoxylsulfate, phenylacetylglutamine, kynurenine, 3-methyl-histidine, trans-4-hydroxyproline, tryptophan, and valine (the last two metabolites differ from the others because their serum concentrations vary directly with GFR).⁴⁹⁻⁵¹

GFR ESTIMATING EQUATIONS

Overview

There are multiple equations to estimate GFR from creatinine, cystatin C, and their combination. Because of variation in the non-GFR determinants across populations, especially for creatinine, it is not possible to use the same eGFR equations across the globe. The 2024 KDIGO Guidelines for CKD recommend use of a single eGFR equation from the limited set of validated equations for each marker in adults and children for use in a region (Table 23.3).³ To facilitate consistency of estimates for patients and comparisons of populations, it would be preferable for each region to be as large as possible, with the understanding the minor variations in accuracy are to be expected within the region. KDIGO recommends that regional disease organizations in collaboration with clinical chemistry groups decide on the appropriate equation for use in the region.

KDIGO identifies criteria for determination of validity of equations that include considerations of the development methods and population, metrics for accuracy compared to measured GFR, and implementation considerations in clinical laboratories. As part of the metrics for accuracy, KDIGO suggests thresholds that are considered sufficient or optimal for use of a GFR estimating equation in practice. As follows, we describe several validated equations for use in adults.

Estimated GFR from creatinine

Before 2021, eGFR equations using serum creatinine (eGFR_{cr}) used variables of age, sex, and race (usually categorized as African American vs. non-African American or

Table 23.3 Most commonly used validated equations for estimating glomerular filtration rate

Marker	Equation name and year	Age	Variables	Development populations
Creatinine	CKD-EPI 2009 ^{191,192}	≥18; Modification CKD-EPI 40 for pediatric available	Developed using ASR but reported not using Black race coefficient, ASR (NB)	8254 Black and non-Black individuals from 10 studies in the United States and Europe ^a
	CKD-EPI 2021 ⁶⁸	≥18	AS	8254 Black and non-Black individuals from 10 studies in the United States and Europe ^a
	EKFC 2021 ⁶	2-100	AS, European Black and non-Black specific Q-value; Separate Q-values for Africa vs. Europe	mGFR vs. SCr (11,251 participants in 7 studies in Europe and 1 study from United States) Normal GFR from 5482 participants in 12 studies of kidney donor candidates 100% Caucasian) European non-Black Q from 83,157 laboratory samples (age 2-40 years) in 3 European hospital clinical laboratories; European Black Q-value (N = 90 living kidney donors from Paris); African Black Q-value (N = 470 healthy individuals from République Démocratique de Congo); All Q-values developed in cohorts independent for EKFC development and validation
	Lund-Malmö Revised 2011 ¹⁹³		AS	3495 GFR examinations from 2847 adults from Sweden referred for measurement of GFR
	CKD-EPI 2009 Modified for China 2014 ^b	≥18	AS	589 People with diabetes from X Third Affiliated Hospital of Sun Yat-sen
	CKD-EPI 2009 Modified for Japan 2016 ^b	≥18	AS	413 Hospitalized Japanese people in 80 medical centers
	CKD-EPI 2009 Modified for Pakistan 2013 ^{b,194}	≥18	AS	542 Randomly selected low- to middle-income communities in Karachi and 39 people from the kidney clinic
	CKiD U25 2021 ⁸³	1-25	AS, height	928 children with CKD in the United States
	CKD-EPI 2012 ¹⁹⁵	≥18	AS	5352 Black and non-Black individuals from 13 studies in the United States and Europe
	CKD-EPI 2023 ⁶³	≥18	A	5352 Black and non-Black individuals from 13 studies in the United States and Europe
Cystatin C	EKFC 2023 ⁵	2-100	A	mGFR vs. SCys (assumed to be the same as mGFR vs. SCr) Normal GFR (same as for SCr equation) Q from laboratory samples from 227,643 (42% female) laboratory samples from Uppsala University Hospital, Sweden
	CAPA 2014 ¹⁹⁶	2-86	AS	4690 Individuals within large subpopulations of children and Asian and Caucasian adults
	CKiD U25 2021 ⁸³	1-25	AS	928 Children with CKD in United States
Creatinine-cystatin C	CKD-EPI 2012 ¹⁹⁵	≥18	Developed using ASR but reported not using Black race coefficient, ASR (NB)	5352 Black and non-Black individuals from 13 studies in the United States and Europe
	CKD-EPI 2021 ⁶⁸	≥18	AS	5352 Black and non-Black individuals from 13 studies in the United States and Europe

^aAlso included 100 Asians and 353 Hispanic or Native Americans.

^bModified from CKD-EPI or MDRD; modifications may reflect systematic differences in measurement of creatinine and mGFR, as well as population differences in nonGFR determinants of creatinine. To obtain eGFR_{cr-cys}, one can average LMR 2011 and CAPA 2014; CKiD U25 creatinine and cystatin C; or EKFC creatinine 2021 and cystatin C 2023 equations.

A, Age; CAPA, Caucasian and Asian pediatric and adult subjects; CKD-EPI, Chronic Kidney Disease Epidemiology Collaboration; CKiD, Chronic kidney disease in children; cr, creatinine; cys, cystatin C; EKFC, European Kidney Function Consortium; GFR, glomerular filtration rate; mGFR, measured glomerular filtration rate; NB, non-Black; R, race; S, sex; SCr, serum creatinine; SCys, serum cystatin C.

From KDIGO. 2024 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease. Stevens, Paul E. et al. *Kidney Int.* 2024;105(4):S117-S314.

Black vs. non-Black) as surrogates for non-GFR determinants of serum creatinine; for example, the MDRD Study 1999 equation recommended by the 2002 KDOQI Guideline and the CKD-EPI 2009 ASR equation recommended in the KDIGO 2012 Guideline for adults in North America, Europe, and Australia. However, race and ethnicity are dynamic, shaped by geographic, cultural, and sociopolitical forces, so the definition can change across geography and over time.^{52,53} It is now recognized that use of a variable for race groups should be avoided in medical algorithms when possible.⁵⁴⁻⁶⁰

These and other considerations led to the 2021 recommendation that race not be used in the computation of eGFR in the United States, and adoption of the CKD-EPI 2021AS equation, which includes only age and sex in addition to creatinine, even though it is slightly less accurate than the 2009 equation in both Black and non-Black populations in the United States.⁶¹ Other countries have also recognized that race should not be included in computation but have elected to continue to use the CKD-EPI 2009 ASR-NB equation, using the non-Black coefficient for all races.⁶² The rationale for this decision in these countries is that the Black population is not large enough to warrant the error in the non-Black population. The European Kidney Function Consortium (EKFC) equation, which uses age, sex, and a Q-value to normalize the serum creatinine to the average value in the population, appears as accurate as the CKD-EPI 2009 ASR-NB equation in the European white population. The latest version of the EKFC recommends race-specific Q-values in multiracial populations, which is not consistent with a race-free equation. The Lund-Malmö Revised 2011 equation, using only age and sex in addition to serum creatinine, appears as accurate as other equations in Sweden. These equations result in similar eGFR values across most of the range of GFR, which affect clinical decision making.⁶³

The non-GFR determinants of age and sex are not sufficient to capture the variation in non-GFR determinants of creatinine across other geographic regions. Several Asian countries (e.g., Japan and Thailand) have developed modifications to the CKD-EPI equation to create region-specific equations.⁶⁴

Estimated GFR from cystatin C

Estimated GFR equations using serum cystatin C (eGFRcys) generally use age and sex or age alone as surrogates for non-GFR determinants of serum cystatin C. The CKD-EPI 2012 equation includes age and sex and was recommended by the KDIGO 2012 Guideline for use in adults. The CAPA 2014 and EKFC 2023 and CKD-EPI 2023 equations include age only and appear as accurate as the CKD-EPI 2012 equation.⁶³

Estimated GFR from creatinine and cystatin C

Multiple studies in adults and children have shown that estimated GFR from the combination of creatinine and cystatin C (eGFRcr-cys) provides more accurate estimates than eGFRcr or eGFRcys alone. The hypothesized rationale is that if the non-GFR determinants of two or more filtration markers are not correlated, then estimated GFR using a combination of the markers is more accurate than estimated GFR using any of the markers alone.⁶⁵ eGFRcr-cys equations include both age and sex. The CKD-EPI 2021 AS equation is now recommended rather than the 2012 ASR equation, which included race. Alternatively, eGFRcr-cys can be computed

using the average of the eGFRcr and eGFRcys using the CKD-EPI 2021 and 2012 equations, Lund-Malmö 2011 and CAPA 2014 equations, or EKFC 2021 and 2023 equations.

Panel estimated glomerular filtration rate

In principle, eGFR from a panel of filtration markers (panel eGFR) can improve accuracy of eGFR and reduce the need for surrogates for non-GFR determinants, and several studies have shown panel eGFR as accurate as or more accurate than eGFRcr-cys without the need for demographic factors.^{51,65} Panels composed of metabolites or low-molecular-weight proteins in addition to creatinine and cystatin C are under investigation but unavailable for clinical use at present time.

Estimated creatinine clearance

The Cockcroft-Gault (CG) equation was developed in 1976 to estimate nonindexed Clcr in adults from age, sex, and body weight in addition to serum creatinine, before standardization of serum creatinine assays. It is not included in the list of validated eGFR equations because multiple studies consistently demonstrate that the CG equation is substantially less accurate than the eGFRcr equations discussed earlier, especially at older age and higher weight. Modifications of the CG equation to account for inaccuracies at larger body size and use with standardized serum creatinine have not been shown to make the CG equation as accurate as eGFRcr equations. The CG equation remains in use today only for dose adjustment of drugs whose pharmacokinetics were evaluated before the availability of eGFRcr equations, but several sources of data suggest that this approach is less accurate than using eGFR.³

APPROACH TO GFR EVALUATION

The recommended approach for evaluation of GFR begins with an initial test, followed by supportive tests when indicated. eGFRcr is the initial test as creatinine is widely available, inexpensive, and ordered routinely.^{1,3,45,66} However, in the steady state in most ambulatory cohorts, accuracy is limited; only 75% to 85% of eGFRcr values are within 30% of mGFR (P_{30}). For clinical settings where greater accuracy is required for clinical decision making or for individuals in whom there are concerns that eGFRcr may be even less accurate, supportive tests are recommended.^{3,5,45,67,68} eGFRcr-cys generally provides the most accurate eGFR (P_{30} as high as 90%) and is recommended as the primary supportive test. For clinical settings or individuals requiring even more accurate GFR assessment for clinical decision making, measured GFR using an exogenous filtration marker is recommended.

This approach requires recognition of the potential for error in estimated and measured GFR (Fig. 23.3). The most important sources of error in eGFR are non-GFR determinants of either serum creatinine or cystatin C. Discordance between eGFRcr and eGFRcys can be used as an indicator of the presence of non-GFR determinants of creatinine and cystatin C. Cross-sectional studies show that eGFRcr-cys is generally more accurate relative to mGFR than either eGFRcys or eGFRcr irrespective of the sign or magnitude of the difference between eGFRcys and eGFRcr (eGFRdiff).^{66,69} There are exceptions, such as in otherwise healthy populations with large deviations in the non-GFR determinants of serum creatinine, but not serum cystatin

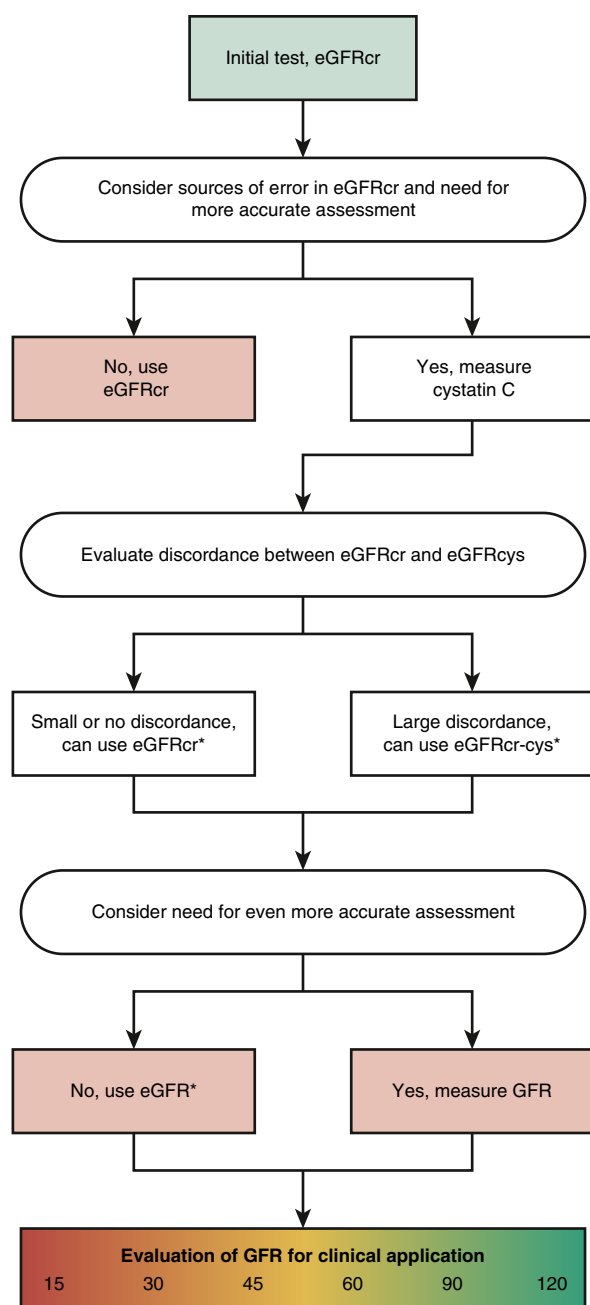


Fig. 23.3 Glomerular filtration rate (GFR) evaluation using initial and supportive tests. The algorithm describes the approach to the evaluation of GFR. Our approach is to use initial and supportive testing to develop a final assessment of true GFR and to apply it in individual decision making at each point in time. The initial test for evaluation of GFR is often *eGFR_{cr}*, which will be available in most patients because creatinine is measured routinely as part of the basic metabolic panel. If the *eGFR_{cr}* is expected to be inaccurate, or if a more accurate assessment of GFR is needed for clinical decision making, such as diagnosis or staging of CKD or drug dosing, then cystatin C should be measured and the discordance between *eGFR_{cr}* and *eGFR_{cys}* should be assessed. If *eGFR_{cr}* and *eGFR_{cys}* are not discordant (within 15 mL/min/1.73 m² or <20%–30% of each other), then accuracy of *eGFR_{cr}*, *eGFR_{cys}*, and *eGFR_{cr-cys}* is similar. If *eGFR_{cr}* and *eGFR_{cys}* are discordant (not within 15 mL/min/1.73 m² or > 20%–30% of each other), then *eGFR_{cr-cys}* is generally more accurate than either *eGFR_{cr}* or *eGFR_{cys}*, with some exceptions, such as otherwise healthy populations with increased creatinine generation owing to increased muscle mass, or decreased creatinine secretion

or extrarenal elimination because of use of specific medications, when *eGFR_{cys}* may be more accurate. If an even more accurate assessment of GFR is needed for a clinical decision, then GFR should be measured using plasma or urinary clearance of exogenous filtration markers, if available. This consideration should be applied any time GFR is required for a clinical decision. It is important to determine how accurate an assessment of GFR needs to be for a clinical decision. P₃₀ for *eGFR* does not generally exceed 90% (90% of *eGFR* within 30% of *mGFR*). P₁₅ for *mGFR* does not generally exceed 90% (90% of *mGFR* within 15% of true *mGFR*). At a GFR of 60 mL/min/1.73 m², 30% accuracy for *eGFR* corresponds to 42 to 78 mL/min/1.73 m², and 15% accuracy for *mGFR* corresponds to 51 to 69 mL/min/1.73 m². At a GFR of 30 mL/min/1.73 m², 30% accuracy for *eGFR* corresponds to 21 to 39 mL/min/1.73 m² and 15% accuracy for *mGFR* corresponds to 26 to 35 mL/min/1.73 m². *Use *eGFR_{cr}* or *eGFR_{cr-cys}* depending on discordance between *eGFR_{cr}* and *eGFR_{cys}*. (Permission granted from *Kidney Med* for Adingwupu OM, Barbosa ER, Palevsky PM. *Kidney Med*. 2023;5(12):100727.)

C, such as decreased creatinine generation due to reduced muscle mass, decreased creatinine secretion, or extrarenal elimination due to use of specific medications. In these cases, *eGFR_{cys}* may be the most accurate.

This approach also requires that the clinician determine how accurate the assessment of GFR needs to be for clinical decision making. If P₃₀ of 75% to 85% is acceptable, then *eGFR_{cr}* may be sufficient, provided there are not large deviations in the non-GFR determinants of serum creatinine. For individuals with expected large deviations in the non-GFR determinants of serum creatinine but not serum cystatin C, *eGFR_{cys}* may be sufficient. If P₃₀ of 90% is acceptable, then *eGFR_{cr-cys}* may be sufficient.

Values for *mGFR* also may often contain an element of error, which differentiates *mGFR* from the “true GFR.” Measurement error is related to the specific filtration marker, the clearance method, and analytical errors in the assay. Studies of repeated measurements within an individual show that only 90% of repeat measurements are within 15% of the original value: P₁₅ of 90%. Thus standardized protocols that accommodate individual patient circumstances, such as reduced level of GFR or volume overload, as well as standardized laboratory measurement procedures, are recommended to ensure the most accurate results.

GFR evaluation in the non-steady state (e.g., during development and recovery from AKI) is less accurate than in the steady state and requires additional considerations. First, as discussed earlier, *eGFR* lags behind true GFR in the non-steady state due to the time required for concentration of the endogenous filtration marker to reach steady state serum concentration; *eGFR* is higher than true GFR when GFR is declining, and *eGFR* is lower than true GFR when GFR is rising. Because of its smaller volume of distribution, serum cystatin C is hypothesized to rise faster than serum creatinine after an acute GFR decline and may be a more sensitive test for the detection of AKI.⁷⁰⁻⁷² Second, changes in fluid status may alter the volume of distribution, with fluid retention leading to slower rise in serum concentrations and fluid depletion leading to faster rise in serum concentrations. Third, the effect of non-GFR determinants may be exaggerated in patients with comorbid conditions that are common in AKI, such as decreased creatinine generation due to muscle wasting, and some studies have demonstrated more accurate drug dosing using algorithms

based on eGFR_{cys} than eGFR_{cr}.⁷³ A kinetic eGFR_{cr} equation has been developed and shown to be predictive of kidney recovery in intensive care unit patients and of delayed graft function in kidney transplant recipients,⁷⁴⁻⁷⁷ but it significantly underestimates mGFR.⁷⁸ More accurate evaluation can be achieved by GFR measurement, although depending on the rapidity of decline in GFR, this test may overestimate GFR.⁷⁹ Alternatively, brief-duration urinary Cl_{cr} can be performed, assuming that the serum creatinine concentration does not increase rapidly over 2 to 8 hours.

GLOMERULAR FILTRATION RATE ASSESSMENT IN SPECIFIC POPULATIONS

The rationale for needing an alternative approach to GFR evaluation in a specific population or disease is that the currently recommended approach for use in the general population is less accurate in that population than in comparable patients in other populations and that an alternative approach is more accurate in that population.^{80,81} For eGFR equations, variation in accuracy across populations and among subgroups of the population is common and reflects variability in the non-GFR determinants of the serum concentrations of the filtration markers associated with these conditions, which is generally greater for eGFR_{cr} than eGFR_{cys}. For children or pregnancy, specific approaches to GFR evaluation are recommended. In other diseases or conditions, such as oncology, equations that are recommended for the general population are recommended for use.⁸² We do not recommend use of disease-specific estimating equations in clinical practice because each patient can have more than one disease, and selecting a particular disease-specific estimating equation would be arbitrary.^{82a}

CHILDREN

Normal values for GFR increase with age since birth, reaching the normal value for adults by approximately 2 years of age. GFR less than 90 mL/min/1.73 m² is generally considered low in children. As in adults, a clearance measurement of an exogenous filtration marker is the reference standard for GFR measurement in children; however, urinary clearances using spontaneous bladder emptying are more difficult in young children, and plasma clearance measurements need to take into consideration the smaller volume of distribution of the marker. Serum creatinine concentrations in children are lower than in adults because of lower muscle mass, so eGFR_{cr} equations developed in adult populations, such as the CKD-EPI equation, are not appropriate for use in children. Multiple eGFR_{cr} equations derived for GFR estimation in children are available, many including height as a variable. The CKiD equation is recommended (Table 23.3).⁸³ An alternative approach may be to use equations that use the full-age spectrum approach such as the EKFC equation. Limitations of CKiD and EKFC equations are that they were derived from cohorts of children with CKD and reduced GFR and validation is required in healthy populations.^{5,6,83,84} Serum cystatin C values are similar in children and young adults, and eGFR_{cys} equations derived in children and adults differ less from each other than eGFR_{cr} equations derived in children and adults.⁵

PREGNANCY

The accuracy of eGFR_{cr} equations has not been well studied in pregnancy, but expected changes in serum creatinine

concentrations during pregnancy are well known. Due to the physiologic increases in effective renal plasma flow and GFR and concomitant hemodilution, serum creatinine initially declines and remains low until the third trimester, at which point it steadily rises, peaks a few weeks postpartum, and then returns to prepregnancy concentrations by 18 weeks postpartum.⁸⁵ Changes in serum cystatin C are reported to follow the same trend.⁸⁶ One international meta-analysis synthesizing the results of 4421 creatinine values in pregnancy proposed that serum creatinine concentration >0.86 mg/dL in the first trimester (85% ULN), 0.81 mg/dL in the second trimester (80% ULN), and 0.87 mg/dL in the third trimester (86% ULN) could be considered abnormal.⁸⁷ Because of lesser variability in sex or age among pregnant women than in the general population, there may be fewer advantages of using eGFR_{cr} equations to translate variations in serum creatinine to variations on the GFR scale. However, use of Scr thresholds suffers from the same concerns with regard to sources of inaccuracy as with use of eGFR_{cr}. Supportive tests for GFR evaluation in pregnancy are also difficult. The risks of measuring GFR using exogenous markers in pregnant patients have not been well defined. The 24-hour urine collection for Cl_{cr} is a commonly used method for confirmation of GFR when there is question about the use of serum creatinine alone despite known errors.⁸⁷

PROTEINURIA

Proteinuria (see Chapter 29) is an important diagnostic and prognostic finding. Guidelines emphasize urine albumin rather than total urine protein as a marker of kidney damage and for the definition and staging of AKD and CKD for the following reasons^{1,3}: Urinary albumin is a marker of glomerular and vascular disease and the principal component of urinary protein in most kidney diseases in adults; assays to measure albumin are more precise and sensitive than assays to measure urine protein and can be standardized, whereas assays for urine protein cannot be standardized; and epidemiologic data demonstrate a strong relationship between albuminuria and clinical outcomes, even at low levels.^{1,3,88,89} This section reviews the normal physiology and measurement issues of total protein, albumin, and other specific proteins found in the urine.

NORMAL PHYSIOLOGY

Approximately 50% of total urine protein is derived from plasma proteins via glomerular filtration and incomplete tubular reabsorption; the remainder is derived from tubular secretion and production by the lower urinary tract and, in men, by the genital tract. Plasma proteins have a wide spectrum of molecular weight and charge and are traditionally classified as albumin (molecular weight [MW] 68,000 daltons, most anionic) and globulins, including γ globulins (immunoglobulins, MW approximately 150,000 daltons, most cationic) and α₁, α₂, and β globulins (MW 15,000–90,000 daltons, intermediate charge). The glomerular filtration barrier acts as a size-, shape-, and charge-dependent permselective molecular sieve, restricting the passage of larger and more anionic macromolecules (MW >20,000 daltons) across the glomerular basement membrane. (See Chapter 4 for additional details.)

Consequently, only about 0.01% or 1 g/day of protein passes into the glomerular filtrate, which is composed primarily of low-MW globulins, including α_2 -microglobulin, β_2 -microglobulin, apoproteins, enzymes, peptide hormones, and immunoglobulin light chains.^{90,91} Filtered serum proteins are broken down by proximal tubular luminal endopeptidases and resorbed primarily via apical endocytic receptors, megalin, and cubilin.^{92,93} Uromodulin (also known as Tamm-Horsfall glycoprotein) is the predominant urine protein not of plasma origin. It is a large glycoprotein (MW 23×10^6 Da) formed on the epithelial surfaces of the thick ascending limb of the loop of Henle and early distal convoluted tubule and secreted into the urine. Tubular secretion of IgA and, in men, proteins produced in seminal, prostatic, and urethral secretions contribute a small percent to total urine protein.

NORMAL VALUES

Urinary protein loss is often referred to as excretion. Normal mean values for urine protein and albumin excretion rates (PER and AER, respectively) are less than 50 and 10 mg/daily, but because of the wide variability, clinical laboratories generally consider the upper limits of normal to be approximately 150 mg and 30 mg/day (Table 23.4). In addition to AKD and CKD, a number of factors influence AER, likely through alterations in glomerular capillary Starling forces and K_F . AER is higher during upright than supine posture and can increase after exercise, during fever, and with weight gain.^{1,3,94} Activation of the renin-angiotensin-aldosterone system and inhibition of tubuloglomerular feedback can increase AER, and medications that interfere with these systems, such as ACE inhibitors, ARBs, MRA, and SGLT2 inhibitors, can lower AER.

Postural proteinuria has been noted in young people without other evidence of kidney disease. It is defined as increased proteinuria with upright posture and normal values with supine posture. It usually does not exceed 1 g total

protein in 24 hours. Characterization of the type of protein has not been reported. Kidney histologic examination generally yields normal or nonspecific findings,^{95,96} and patients with postural proteinuria have not been observed to develop kidney disease.⁹⁷ Hypothesized mechanisms are that this is a normal variant or that it reflects a subtle glomerular abnormality, an exaggerated hemodynamic response, or left renal vein entrapment. As discussed next, the finding of proteinuria in a young person should prompt a testing of an early morning specimen to exclude postural proteinuria.⁹⁸

PATTERNS OF PROTEINURIA IN DISEASE

GLOMERULAR PROTEINURIA

Damage to the glomerular filtration barrier causes loss of charge and size selectivity, leading to increased filtration and excretion of high-MW and anionic proteins, in particular albumin. Because albumin is the dominant high-MW protein in plasma, it is therefore the dominant protein in urine in glomerular diseases.^{91,99}

TUBULAR PROTEINURIA

Tubular damage or dysfunction may reduce the normal reabsorptive capacity of the proximal tubule, resulting in increased excretion of low-MW proteins in the urine. Classic causes of tubular proteinuria are congenital tubulointerstitial diseases, such as Fanconi syndrome, Imerslund-Gräsbeck syndrome, or Dent disease; secondary causes of Fanconi syndrome include systemic diseases, drugs, and toxins. In addition, increased excretion of low-MW proteins may occur in glomerular disease, likely reflecting the coexistence of tubular interstitial disease.

OVERFLOW PROTEINURIA

Increased production of normal or abnormal plasma low-MW proteins can cause increased filtration at the glomerulus and overwhelm the normal reabsorptive

Table 23.4 Kidney disease: Improving global outcomes (KDIGO) guideline: categories of proteinuria

	Normal to mildly increased (A1)	Moderately increased (A2)	Severely increased (A3)
Timed urine collection			
AER (mg/24 h)	<30	30-300	>300
PER (mg/24 h)	<150	150-500	>500
Spot or timed collection			
ACR:			
mg/mmol	<3	3-30	>30
mg/g	<30	30-300	>300
PCR:			
mg/mmol	<15	15-50	>50
mg/g	<150	150-500	>500
Protein reagent strip	Negative to trace	Trace to +	+ or greater

Relationships between AER and ACR and between PER and PCR are based on the assumption that average creatinine excretion rate is 1.0 g/day or 10 mmol/day (conversions are rounded for pragmatic reasons). ACR and PCR may also be reported as $\mu\text{g}/\text{mg}$ ($1 \mu\text{g}/\text{mg} = 1 \text{ mg}/\text{g}$) or mg/mg ($1 \text{ mg}/\text{mg} = 0.001 \text{ mg}/\text{g}$).

ACR, Albumin-to-creatinine ratio; AER, albumin excretion rate; PCR, protein-to-creatinine ratio; PER, protein excretion rate.

From Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group. KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int Suppl*. 2013;3:1-150.

capacity of the proximal tubule, leading to increased excretion in the urine. This occurs particularly with small or positively charged proteins and is of clinical importance principally in monoclonal gammopathies due to plasma cell or B-lymphocyte disorders, in which immunoglobulin light chains (MW 25,000 daltons) are increased in serum and urine (Bence Jones proteins). It may also occur due to release of myoglobin in rhabdomyolysis or hemoglobin in severe intravascular hemolysis.

PROTEINURIA OF URINARY TRACT ORIGIN

Mildly increased excretion of albumin and other high-MW proteins may occur in the setting of urinary tract infections or stones, reflecting exudation of protein into the urine. Leukocytes are also commonly present in the urine sediment in these settings.

MEASUREMENT OF PROTEINURIA

For types of urine specimens and techniques for obtaining them, see the following section on urine dipstick.

ASSAYS

Assays for total urine protein and albumin are the most commonly used assays for urine protein. Assays are available for other proteins, such as specific low-MW proteins and immunoglobulin light chains.

Total urine protein is primarily assayed using turbidimetric methods after precipitation by a strong acid. Because of the wide spectrum of proteins in urine in normal and pathologic states, there cannot be a standardized reference material or reference measurement procedure for urinary total protein. Each measurement method has a different lower limit of quantification and different sensitivity and specificity for threshold concentrations for the diverse range of proteins found in urine.¹⁰⁰⁻¹⁰²

Urine albumin is generally measured by turbidimetric or nephelometric immunoassay with typical lower limits of quantification being approximately 5 mg/L.¹⁰³ Thus relatively large increases in urinary albumin excretion can occur without causing a measurable increase in urinary total protein.¹⁰⁴ Polyclonal and monoclonal antibodies are used, and different antibodies target different epitopes, which can be a source of variability across assays. The most common reference material used is RMM-ERM DA470 (CRM 470); standardization of the reference material and reference measurement procedure is in progress.^{105,106} Sources of error in immunoassays of urine albumin include high urine albumin concentration, causing a high-dose “hook effect” phenomenon, which can be seen in a one-step immunometric or sandwich immunoassay due to saturation of capture and detection antibodies by an excess of antigens.¹⁰⁷

TIMED VERSUS UNTIMED RANDOM COLLECTION

Urine protein or urine albumin measurement in 24-hour urine collection is generally considered the gold standard for PER and AER but inconvenient and difficult to supervise, and errors due to inaccurate timing and incomplete bladder emptying are common. Shorter duration of timed urine collections can reduce inconvenience and errors in timing

but must take into account diurnal variation in PER and AER.

Untimed “spot” specimens can be collected at a specified or random time, expressed as a concentration. They can overcome the disadvantage of timed collections but are not reliable for estimation of PER and AER because of the variation in urine concentration.

URINE PROTEIN OR ALBUMIN-CREATININE RATIO

To overcome the limitations of variation in urine concentration of protein and albumin in untimed urine collections, the concentrations can be divided by the creatinine concentration to yield the protein- and albumin-creatinine ratio (PCR or ACR), expressed as mg/g or mg/mmol. Concentration ratios are more accurate and have less intraindividual variability than concentrations in untimed samples.¹⁰⁸⁻¹¹⁰ Under normal circumstances, creatinine excretion rate (CER) is relatively constant throughout the day; thus diurnal variation in PCR and ACR reflects diurnal variation and PER and AER.¹¹¹ A first-morning void sample has the least intraindividual variability^{1,112,113} and is required to exclude the diagnosis of orthostatic (postural) proteinuria.^{114,115} A random urine sample is acceptable if a first-morning void sample is not available.

Urine concentration ratios are affected by within-person variation in CER, as well as by population variation in PER and AER. In the steady state, CER reflects muscle mass and dietary intake, and on average it is higher in men, younger people, more muscular individuals, and those with a higher meat intake and lower in women, older people, people with malnutrition or muscle wasting, and vegetarians.^{116,117} Thus average values for PCR and ACR in the population are lower for men than women and increase with age.^{118,119} Changes in GFR lead to a non-steady state of urine CER. A decline in GFR leads to decreased CER and increased PCR and ACR, while a rise in GFR leads to increased CER and decreased PCR and ACR.

Twenty-four-hour PER and AER are often estimated from PCR and ACR by assuming that 24-hour CER is 1000 mg or 10 mmol. Because of variation of CER, this assumption provides only a rough approximation of PER and AER. CER can be estimated from age, sex, and other factors, which can be used to convert PCR to PER and ACR to AER, but the accuracy of these estimates is limited.^{120,121}

DIPSTICK

Dipstick urine protein testing is used widely, as a single reagent dipstick or as part of a multireagent dipstick (for more on multireagent dipsticks, see the following section on urinalysis). The lower detection limit for reagent methods is 150 to 300 mg/L, and the test is more sensitive to albumin than to globulins, immunoglobulin light chains, and hemoglobin. Reagent strips are also available for detection of albuminuria on the basis of photometric and immunologic methodologies.¹²² Most dipstick reagents for urine protein are semiquantitative, containing a pH-sensitive colorimetric indicator that changes color when binding negatively charged proteins. Dipstick testing has greater sensitivity for albumin than positively charged, nonalbumin proteins¹²³⁻¹²⁶ and therefore may have false-negative results in the presence of predominantly tubular or overflow proteinuria.¹²⁷ Albumin-specific dipsticks that are

more accurate than usual dipsticks are available, but they are less accurate than quantitative immunoassays discussed earlier.¹²⁸

All dipstick tests are strongly affected by urine concentration. Very dilute urine may give false-negative results, and concentrated urine may give false-positive results. A very high urine pH (>7.0) can give false-positive results, as can contamination of urine with blood. Thus multiple reagent dipsticks that measure specific gravity, pH, and heme concurrently with urinary protein can help interpret a protein result.¹²⁹ Interoperator variability can be reduced by use of an automated reader.^{112,125}

APPROACH TO PROTEINURIA EVALUATION

The recommended approach for evaluation of proteinuria in adults begins with an initial test, followed by confirmatory tests when indicated (Table 23.5). Spot urine ACR is the initial test in adults, but there is a hierarchy of tests if ACR is not available as an initial test, with subsequent confirmation by ACR. AER in a timed urine sample is the generally accepted confirmatory test for urine ACR but is performed only when urine ACR is not accurate enough for clinical decision making.³ Equations are available to convert measures of proteinuria to ACR or AER, but accuracy is limited at the lower range (ACR <500 mg/day).¹³⁰⁻¹³²

As discussed earlier, measurement of urine albumin and not urine protein is recommended in adults.^{1,3} However, this approach may overlook “tubular” and “overflow” proteinuria, in which nonalbumin proteins predominate. If these conditions are suspected, they may be detected by measurement of total urine protein in addition to urine albumin; increased level of total urine protein with albumin-protein ratio of <0.4 suggests presence of an increased

Table 23.5 Evaluation of proteinuria in adults

- Initial testing of proteinuria (in descending order of preference, in all cases an early morning urine sample is preferred)
- Urine albumin-to-creatinine ratio (ACR)
 - Urine protein-to-creatinine ratio (PCR)
 - Reagent strip urinalysis for total protein with automated reading
 - Reagent strip urinalysis for total protein with manual reading
- Confirmatory testing for proteinuria
- Confirm reagent strip positive albuminuria and proteinuria by quantitative laboratory measurement and express as a ratio to creatinine wherever possible.
 - Confirm ACR ≥ 30 mg/g on a random untimed urine with a subsequent early morning urine sample.
 - If a more accurate estimate of albuminuria or total proteinuria is required, measure AER or PER in a timed urine sample

Clinical laboratory reporting of ACR and PCR in untimed urine samples in addition to albumin concentration or proteinuria concentrations rather than concentrations alone.

From Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group. KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int Suppl.* 2013;3:1–150.

level of a nonalbumin protein.¹³³ Confirmation of tubular proteinuria can be obtained by immunoassays directed at specific low-MW proteins in urine. Confirmation of immunoglobulin light chains is best obtained by measuring serum assays for immunoglobulin light chains.¹³⁴

This approach requires recognition of the potential for errors in measurement of urine albumin and total urine protein (Table 23.6). Because of substantial day-to-day biological variability in urine albumin and urine protein, a positive result should be followed with a second measurement, ideally in an early morning sample, to confirm the result.^{1,3}

PROTEINURIA MEASUREMENT IN SPECIFIC POPULATIONS

CHILDREN

PCR is the preferred initial test for children as the majority of children have underlying developmental abnormalities often referred to as congenital anomalies of the kidney and urinary tract (CAKUT), and a much higher proportion of children than adults have tubulointerstitial diseases with tubular proteinuria.¹³⁵ With the rising prevalence of obesity and type 2 diabetes in children, there is growing

Table 23.6 Biological, preanalytical, and analytical factors that may impact variability in urinary protein or albumin or ACR

Biological

Increase in albumin or protein	Hematuria High-intensity exercise Uterine bleeding Posture (postural proteinuria)
Increase in protein	Urinary tract infection or inflammation
Increase in ACR or PCR due to lower urine creatinine	Older age Female sex Reduced muscle mass Decreased dietary protein intake Decreased physical activity

Preanalytical

Collection type	Timed or untimed (spot)
Time of day	First morning vs. random void (first morning reduces variation)
Storage conditions	Degradation of protein or albumin during storage Adsorption to plastic Storage temperature

Analytical

Total protein or albumin measurement
High albumin levels can result in high-dose hook effect

Adapted from Johnson DW, Jones GR, Mathew TH, et al. Chronic kidney disease and measurement of albuminuria or proteinuria: a position statement. *Med J Aust.* 2012;197:224–225; and Miller WG, Bruns DE, Hortin GL, et al. Current issues in measurement and reporting of urinary albumin excretion [article in French]. *Ann Biol Clin (Paris).* 2010;68:9–25.

concern that relying on PCR may overlook the early onset of albuminuria, and guidelines recommend ACR as the initial test in children with diabetes.³

PER and AER in children are indexed by BSA; normal PER is <10 mg/m²/day, or <4 mg/m²/hour.¹³⁶ Nephrotic-range proteinuria is defined as 1000 mg/m²/day, or 40 mg/m²/hour, or higher. Urine PCR has been shown to have reasonable accuracy in reflecting 24-hour PER in a small study of 15 children.¹³⁷ As in adults, higher PCR has been shown to predict an increased rate of GFR decline in children.¹³⁸⁻¹⁴⁰ Normal mean AER in children is approximately 10 mg/1.73 m^{2.3} Use of ACR threshold of 30 mg/g has been recommended for detection of albuminuria.

PREGNANCY

Dipstick protein is used as a screening test for preeclampsia, with PER as a confirmatory test, with a threshold of ≥300 mg/day. The roughly equivalent urine PCR of 300 mg/g (30 mg/mmol) showed reasonable performance in estimating 24-hour protein excretion and as a rule-out test in two systematic reviews of studies in this setting, although there were no data on PCR for predicting outcomes. Currently, total protein remains the recommended method used to test for suspected preeclampsia, but urine albumin is recommended if glomerular disease is suspected.^{141,142}

URINE DIPSTICK

The dipstick, a multireagent test strip with colorimetric assays for a variety of urine properties, is the initial test for many chemical analyses or cytologic analyses of the urine. Dipstick testing can be either qualitative, denoting the presence or absence of the substrate, or semiquantitative, giving a crude estimate of the quantity of the substrate, with manual or automated reading. On the basis of the results of the dipstick, the urine can be sent for confirmatory testing for more precise quantification.

PREPARATION AND METHOD

Urine specimens for dipstick and microscopy can be obtained by spontaneous voiding, urethral catheterization, and percutaneous bladder puncture. Technique is important in collecting a sample to avoid contamination. For spontaneously voided urine, a midstream sample should be collected after cleaning of the external genitalia. If a patient has an indwelling catheter, a fresh specimen should be submitted for analysis; samples that have been stagnant in the catheter tubing or bag may have undergone degradation. Suprapubic needle aspiration of the bladder is used when urine cannot easily be obtained by other means, most commonly in infants. Whatever the collection method, it is recommended that a sample be analyzed within 2 to 4 hours of the time of collection to avoid precipitation of solutes and cell lysis.¹²³

CHEMICAL CONSTITUENTS

COLOR

The color of urine is determined by chemical content, concentration, and pH (Table 23.7). Urobilin, a degradation product of heme, is primarily responsible for the yellow color of urine (“urochrome”). Urine may be dark yellow in concentrated urine and almost colorless in dilute urine. Abnormal color can be due to drugs, foods, and pathologic conditions. Cloudy urine is most commonly due to leukocytes and bacteria. The most common cause of red urine is hemoglobin. Pink or orange urine may be due to ingestion of pigment with medications (rifampin or phenazopyridine), foods (beets), or some porphyrias.

ODOR

The odor of normal urine is due to ammonia, a breakdown product of urea, and is stronger in concentrated urine and weaker in dilute urine. The most common cause of abnormal urine odor is infection; the abnormal odor is caused by the production of ammonia by bacteria that

Table 23.7 Main causes of abnormal color changes in urine		
	Cause	Color
Pathologic conditions	Macroscopic hematuria, hemoglobinuria, myoglobinuria	Pink, red, brown, black
	Jaundice	Yellow to brown
	Chyluria	White milky
	Massive uric acid crystalluria	Pink
	Porphyria, alkaptonuria	Red to black; increases after urine left to stand
Medications	Rifampin	Yellow-orange to red
	Propofol	White
	Phenytoin, phenazopyridine	Red
	Chloroquine, nitrofurantoin	Brown
	Triamterene, blue dyes of enteral feeds	Green
	Metronidazole, methyldopa, imipenem–cilastatin	Darkening after urine left to stand
Foods	Beetroot	Red
	Senna rhubarb	Yellow to brown-red
From Fogazzi GB, Verdesca S, Garigali G. Urinalysis: core curriculum 2008. <i>Am J Kidney Dis.</i> 2008;51:1052–1067; and Davsion A. <i>Urinalysis</i> . 3rd ed. Oxford: Oxford University Press; 2005.		

produce the enzyme urease. Ketones may cause a fruity or sweet odor. Other rare pathologic conditions include maple syrup urine disease (maple syrup odor), phenylketonuria (mousy odor), isovaleric acidemia (sweaty feet odor), and hypermethioninemia (fishy odor).¹²³

CONCENTRATION

Urine concentration reflects the activity of the distal nephron to concentrate or dilute the urine; deviations from the expected value may be a marker of kidney damage or may occur in systemic disorders (see also Chapter 10). Urine osmolality, defined as the number of osmoles per kilogram, is the reference standard for assessment of urine concentration. The physiologic range for urine osmolality is from approximately 50 to 1400 mOsm/kg (the normal plasma osmolality is approximately 290 mOsm/kg; 0 mOsm/kg is the value for water). Relative density is assessed by specific gravity, which is defined as the weight of a solution relative to that of an equal volume of water. The physiologic range for urine-specific gravity is 1.001 to 1.040; 1.000 is the value for water. Urine-specific gravity is generally proportional to the osmolality. A specific gravity of approximately 1.010 is associated with urine osmolality of 290 mOsmol/kg and is approximately 0.001 unit higher or lower for every 35 to 40 mOsmol/kg higher or lower urine osmolality, respectively.¹⁴³

Specific gravity is traditionally measured by a hydrometer (the depth of flotation is inversely related to the density of urine) or refractometer (the refractive index is proportional to urine concentration). Reagent strips contain a polyionic polymer with binding sites that are saturated with hydrogen ions and an indicator substance. The release of hydrogen ions when they are competitively replaced with urinary cations causes a change in the pH-sensitive indicator dye that is quantitatively related to the concentration of cations. Specific gravity values measured by dipstick tend to be falsely high if the urine pH is <6 and falsely low if the pH is >7.¹²³ Large-MW molecules such as glucose, protein, and iodinated contrast elevate specific gravity but have a smaller effect on osmolality.^{144,145}

The reference method procedure for urine concentration is the osmometer, most commonly using the freezing point depression method. The freezing point of a solution is depressed by the addition of solute. Osmometers that use this method compare the freezing point of the solution of interest to a reference solution in order to determine the osmolality of the solution.¹⁴⁶

URINE PH

Urine pH reflects the activity of the distal nephron to acidify the urine; deviations from the expected value may be a marker of kidney damage or may occur in systemic disorders (see also Chapter 9). The normal range for urine pH is 4.5 to 7.8. A highly alkaline urine (pH >7.0) is suggestive of infection with a urea-splitting organism, such as *Proteus mirabilis*. Prolonged storage can also lead to overgrowth of urea-splitting bacteria.¹⁴⁷ The reagent test strip measures urine pH using two indicators, methyl red and bromothymol blue, to give a broad range of colors at different pH values. Significant deviations from true pH occur with values <5.5 or >7.5.^{148,149} The reference measurement procedure for

urine pH is the pH meter, with urine collected under oil to prevent alkalization of the urine by evaporation of dissolved carbon dioxide.

BILIRUBIN AND UROBILINOGEN

Only conjugated bilirubin is passed into the urine; a positive result of a reagent test for bilirubin typically occurs in patients with obstructive or hepatocellular jaundice, whereas it is usually negative in patients with jaundice due to hemolysis. Prolonged storage and exposure to light can lead to false-negative results. A false-positive test result for urine bilirubin can occur if urine is contaminated with stool.¹⁴⁷ Urobilinogen is a breakdown product of heme; its presence in the urine suggests hemolysis.

LEUKOCYTE ESTERASE AND NITRITES

Leukocyte esterase is a marker of granulocytes in the urine. The esterase method relies on detection of esterases released from lysed urine granulocytes. These esterases liberate 3-hydroxy-5-phenylpyrrole after substrate hydrolysis. The pyrrole reacts with a diazonium salt in the test strip, yielding a pink to purple color. The result is usually interpreted as negative, trace, small, moderate, or large. Factors that may increase granulocyte lysis include allowing urine to stand for long periods, low pH, and low relative density. In these settings, there may be a positive dipstick result for leukocyte esterase with no leukocytes seen on microscopy. High levels of glucose, albumin, ascorbic acid, tetracycline, cephalixin, cephalothin, or large amounts of oxalic acid may inhibit the reaction and cause false-negative results.¹⁵⁰

Urinary nitrites indicate the presence of nitrate-reducing bacteria. In the reagent strip test, nitrite reacts with a *p*-arsanilic acid to form a diazonium compound; further reaction with 1,2,3,4-tetrahydrobenzo(h)quinolin-3-ol results in a pink color endpoint. Results are usually interpreted as positive or negative. High specific gravity and ascorbic acid may interfere with the test. False-negative results are common and may be due to prolonged sample storage or low dietary intake. It may take up to 4 hours to convert nitrate to nitrite, so short bladder retention time can also give false-negative results.¹⁵⁰

GLUCOSE

Glycosuria indicates hyperglycemia (blood glucose >180 mg/dL [10 mmol/L]) in subjects with normal levels of GFR or failure of proximal renal tubular reabsorption in tubular disorders such as Fanconi syndrome. SGLT-2 inhibitors block the reabsorption of glucose at the proximal convoluted tubule and are a frequent cause of glycosuria.¹⁵¹ Most reagent strips use an oxidase-peroxidase method to measure glucose in the urine. Glucose is first oxidized to form glucuronic acid and hydrogen peroxide. Hydrogen peroxide then reacts via a peroxidase with a reduced chromogen to form a colored product.¹²³ This test is sensitive to glucose concentrations between 0.5 and 20 g/L.¹²³ Large quantities of ketones, ascorbate, and phenazopyridine metabolites may interfere with the color reaction, causing false-negative results. Oxidizing agents and hydrochloric acid may cause false-positive results.¹⁵⁰ Enzymatic methods such as a hexokinase give more precise quantification of urinary glucose levels.¹²³

KETONES

Ketonuria indicates increased ketone production and can occur with elevated serum ketone concentration in alcoholic or diabetic ketoacidosis or with normal serum ketone concentration during prolonged fasting or starvation. Acetoacetate and acetone are generally detected with the nitroprusside reaction.¹⁵⁰ Ascorbic acid and phenazopyridine can give false-positive reactions. β -hydroxybutyrate (which often represents 80% of total serum ketones in ketosis) is not normally detected by the nitroprusside reaction, so serum assays are used to detect β -hydroxybutyrate.

HEMOGLOBIN AND MYOGLOBIN

Heme (MW 616 daltons, a component of hemoglobin and myoglobin) in urine can be due to increased heme filtration due to increased production (hemolysis or rhabdomyolysis) or bleeding into the urinary tract. Normally, haptoglobin binds heme-containing substances, preventing their filtration. When these substances are produced in large quantities, the capacity for binding is overwhelmed and they appear in the urine. Reagent strips are sensitive for the presence of heme in the urine; even small amounts of red blood cells (RBCs) will cause a positive test. A positive dipstick test in the absence of RBCs in the urine sediment therefore suggests either hemolysis or rhabdomyolysis. Reagent strips use the peroxidase-like activity of the heme moiety to catalyze the reaction between a peroxide and a chromogen, giving a colored product. False-negative results are uncommon but may be caused by ascorbic acid, a strong reducing agent. False-positive results may occur because of oxidizing contaminants, povidone-iodine, semen, or a high concentration of bacteria with pseudoperoxidase activity (such as *Enterobacteriaceae*, staphylococci, and *Streptococcus* species).¹²³

URINE MICROSCOPY

Examination of the urine by microscopy has been performed systematically at least since the 19th century¹⁵² and remains a vital component of the laboratory assessment of AKD/CKD. For example, as described later, identification of dysmorphic RBCs and RBC casts in a patient with AKI strongly suggests the diagnosis of rapidly progressive glomerulonephritis and may prompt empirical treatment with immunosuppressive therapy in advance of results of serology or kidney biopsy.

Urine microscopy can be performed by the clinician, clinical laboratory technician, or automated cell counter. Studies have shown that interpretation of urinalysis by a nephrologist is more likely to correctly diagnose AKI than with laboratory analysis alone.^{153,154} However, urine microscopy depends on expertise and has poor interobserver reliability.

PREPARATION AND METHOD

Formed elements in urine degrade rapidly and so urine microscopy is best performed on a fresh urine sample within 2 hours of collection.¹²³ The first-morning specimen

has been recommended in the past for urine microscopy because the urine is acidic and concentrated, but a midstream specimen of the second morning specimen is favored owing to absence of lysis of urine particles after prolonged storage of urine in the bladder overnight. Specimens should preferably not be refrigerated because this may cause precipitation of crystals. High urine pH and dilute urine lead to more degradation of formed elements.

A guideline provided by the European Federation of Laboratory Medicine suggests the following method for preparation of the urine sediment for microscopy.¹⁵⁵ A sample of 5–12 mL of urine should be centrifuged for 5 minutes at a rate per minute appropriate for the size of the centrifuge, a defined volume of supernatant removed by suction rather than arbitrary decanting, and the pellet resuspended by gentle agitation. A drop of urine should be placed on a slide under a coverslip, and it should be examined ideally with phase-contrast microscopy, as well as usual brightfield microscopy, at low power ($\times 100$) and then at high power ($\times 400$). Staining or polarized light may also be useful to identify certain substances.

CELLS OF THE URINARY SEDIMENT

ERYTHROCYTES

The normal upper limit for the number of RBCs in midstream urine is 1 to 2 per high-power field (HPF) by semiquantitative techniques and 8000 RBCs/mL using the counting chamber.¹⁵⁶ “Hematuria” is defined as 3 or more RBCs per high-power field.¹⁵⁷ Hematuria that is visible as red discoloration of urine is classified as “macroscopic hematuria,” while hematuria that is only identifiable via microscopy is classified as “microscopic hematuria.” There is substantial overlap among the causes of macroscopic and microscopic hematuria.

Blood in the urine can arise from multiple locations in the genitourinary tract including the kidney, ureters, bladder, and urethra (Table 23.8). Glomerular hematuria can be distinguished from nonglomerular hematuria by the size and shape of RBCs. Isomorphic RBCs, similar in size to the erythrocytes found in the bloodstream, are thought to be nonglomerular in origin (Fig. 23.4). Dysmorphic RBCs are erythrocytes from the glomerular capillary, and their irregular appearance is a consequence of damage due to pH and osmolality changes as the cells travel through the tubule. Acanthocytes are a subtype of dysmorphic RBCs with protruding blebs. Glomerular hematuria is defined by some institutions in terms of percentage of dysmorphic RBCs, but the threshold at which this value is believed to be significant is not standardized. A common definition of glomerular hematuria is defined as $\geq 40\%$ dysmorphic RBCs or $\geq 5\%$ acanthocytes.¹⁵⁸

Automated methods of examining for glomerular or nonglomerular hematuria using measurements of mean corpuscular volume (MCV) of the red cells have been developed in an attempt to overcome the problems with reliability and reproducibility of urine microscopy. An MCV smaller or larger than the normal range (50–80 fL) is recorded as dysmorphic. A meta-analysis concluded that the

Table 23.8 Causes of hematuria

Location in the urinary tract	Clinical entity	Urine microscopy findings				
		RBC casts or dysmorphic RBCs	WBC	Crystals	Albuminuria	Symptoms
Kidney Glomeruli	Primary glomerular disease (e.g., Alport syndrome, IgA, C3, and dense deposit disease)	+	–	–	±	–
	Secondary glomerular disease (e.g., lupus glomerulonephritis, vasculitis, and postinfectious glomerulonephritis)	+	+	–	+	+
Tubules, interstitium, vessels, cysts	Papillary necrosis	–	±	–	±	+
	Sickle cell disease	–	±	–	±	±
	Autosomal dominant polycystic kidney disease	–	±	–	±	±
	Infarct (thrombosis, embolisms, dissection)	–	±	–	±	±
Upper urinary tract	Benign kidney masses	–	±	–	–	–
	Malignancy	–	±	–	–	±
	Pyelonephritis	–	+	–	–	+
	Nephrolithiasis	–	–	–	–	–
Lower urinary tract	Malignancy (bladder, ureter, prostate)	–	±	–	–	±
	Ureterolithiasis	–	+	±	–	±
	BPH	–	–	–	–	±
	Urinary tract infection	–	+	–	–	+
Any region of the urinary tract	Exercise hematuria	±	+	–	±	±
	Bleeding diathesis or anticoagulation	–	–	–	–	±
	Hypercalciuria and hyperuricosuria	–	±	±	–	–
	Drugs (cyclophosphamide or crystal-inducing drugs)	–	–	±	–	±
	Trauma	–	–	–	–	+

D, Dysmorphic; I, isomorphic; + almost always present; ± sometimes present; – rarely present.

diagnostic value of this test is limited and that the urinary MCV test was not reliable in cases of low-grade hematuria because of interfering debris.¹⁵⁹

LEUKOCYTES

The upper limit of normal for the number of white blood cells (WBCs) in midstream urine is 1 to 2 per HPF by semiquantitative techniques¹⁶⁰ and 2000 to 8000 WBC/mL by counting chamber.^{161,162} The most common leukocytes found in the urine, neutrophils, are usually an indication of inflammation, most commonly urinary tract infection or contamination from genital tract infection. Sterile pyuria should prompt consideration of mycobacterial infection of the urinary tract and noninfectious interstitial nephritis from drug toxicity or allergic or autoimmune diseases. Eosinophils are detectable with Wright stain or Hansel stain. Hansel stain has improved sensitivity, especially because a urine pH of <7 inhibits Wright stain. Although

eosinophiluria was initially associated with drug-induced hypersensitivity, the list of diseases that may be associated with eosinophiluria is diverse and includes renal cholesterol embolism, rapidly progressive glomerulonephritis, and prostatitis and it is no longer thought to be a sensitive test for the diagnosis of allergic interstitial nephritis.¹⁶³ The diagnostic value of the presence of other leukocytes, such as lymphocytes and macrophages, is currently limited, although lymphocytes have been indicative of transplant rejection and macrophages may be found in glomerulonephritis.

RENAL TUBULAR EPITHELIAL CELLS

Renal tubular epithelial (RTE) cells indicate tubular injury and may be present in any type of acute or chronic kidney disease. Depending on the cause of kidney disease, RTE may be accompanied by RBCs or WBCs. If RTE cells are the only cell type and are found in abundance in AKI,

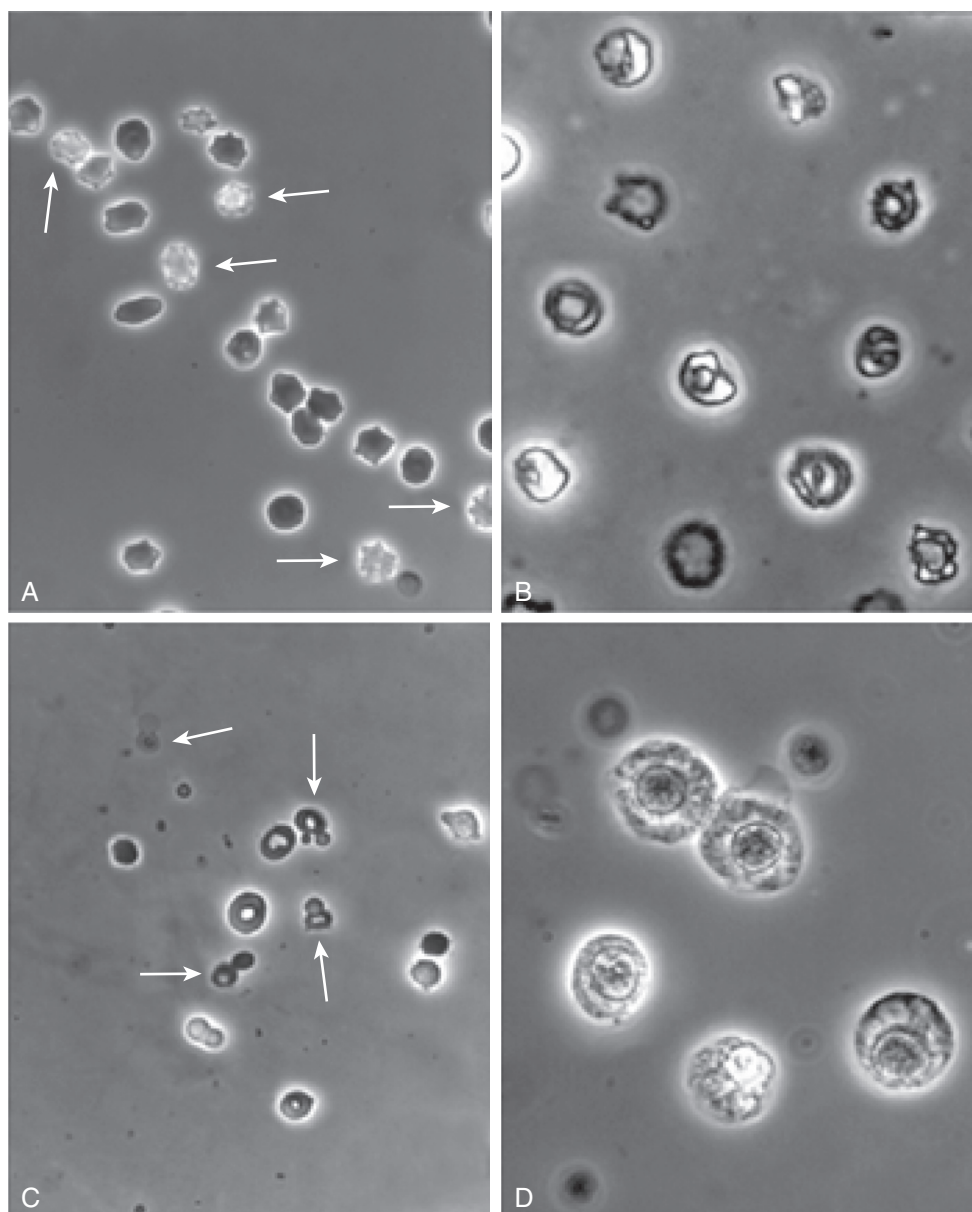


Fig. 23.4 Erythrocytes in urine. (A) Isomorphic erythrocytes, some with a “crenated” appearance (arrows). (B) Different types of dysmorphic erythrocytes. (C) Acanthocytes (arrows). (D) Proximal renal tubule cells. (From Fogazzi GB, Verdesca S, Garigali G. Urinalysis: core curriculum 2008. *Am J Kidney Dis.* 2008;51:1052–1067.)

acute tubular necrosis should be considered. Extracellular coarse granules may reflect debris from necrosis of renal tubular cells.

OVAL FAT BODIES AND LIPIDS

Lipids appear as spherical, translucent drops of varying size within the cytoplasm of RTE cells as “oval fat bodies” or as extracellular elements. Under polarized light, lipid droplets look like Maltese crosses. Lipiduria is usually associated with heavy proteinuria but may also be present in Fabry disease.

OTHER CELLS

Epithelial cells from the lower urinary tract are larger than RTE cells. Squamous epithelial cells, the largest cells of the urinary sediment, derive from the urethra or

external genitalia and indicate contamination of the urine specimen. Bladder epithelial cells (“urothelial cells”) may be seen in urologic diseases such as malignancy.

OTHER ELEMENTS OF THE URINARY SEDIMENT

CASTS

“Casts” (Fig. 23.5) are cylindrical bodies originating in the kidney. They form from the aggregation within the tubules of fibrils of uromodulin. Trapping of various particles within the cast matrix, as well as degenerative processes, results in casts with different appearances and clinical significance (Table 23.9). The findings of specific cell types within a cast indicate kidney disease and have important diagnostic significance about the cause of kidney disease. The term “active urinary sediment” usually refers to the

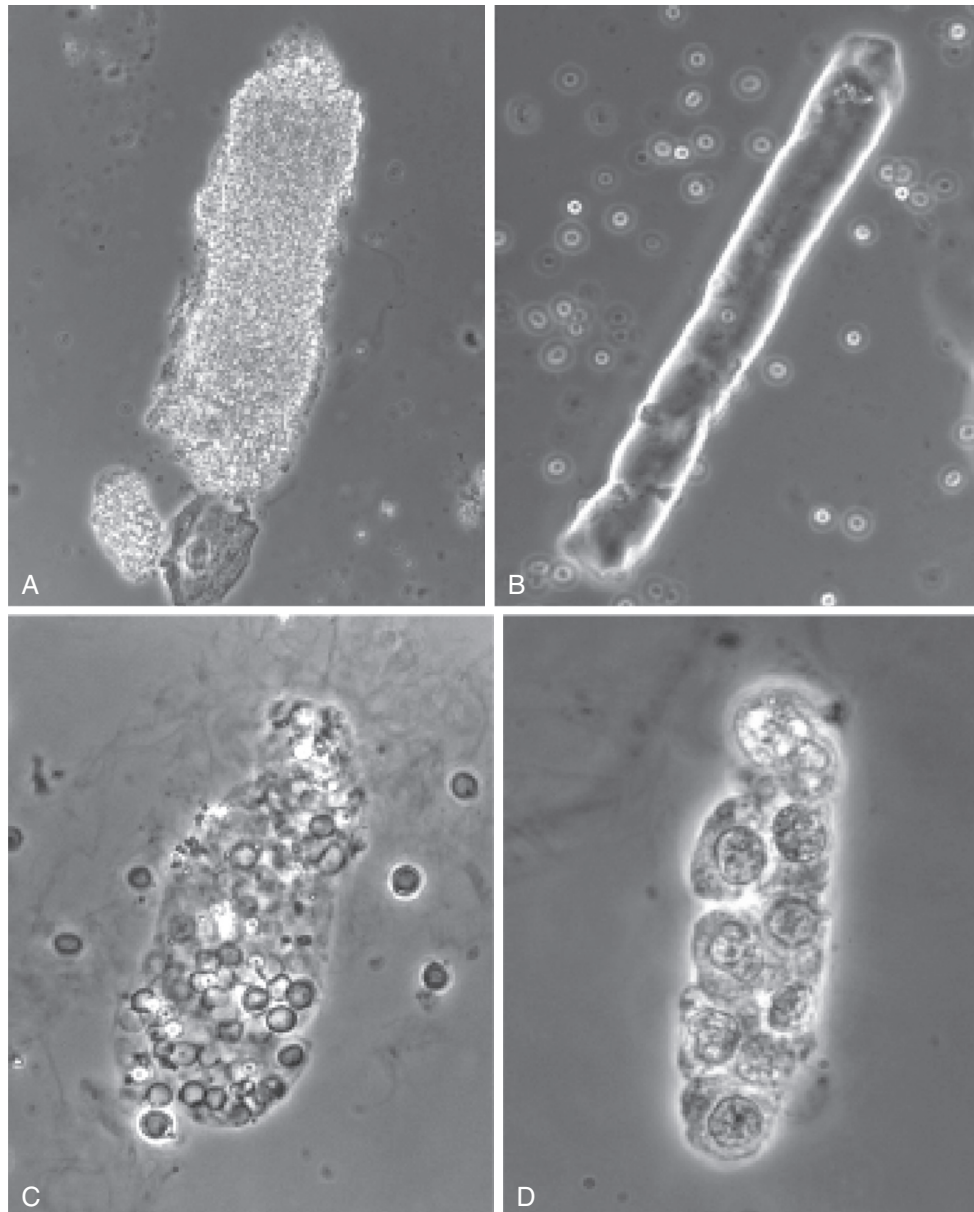


Fig. 23.5 Urinary casts. (A) Finely granular cast. (B) Waxy cast with the typical “melted wax” appearance. (C) Red blood cell cast. (D) Renal tubule epithelial cell cast. (From Stevens LA, Nolin TD, Richardson MM, et al. Comparison of drug dosing recommendations based on measured GFR and kidney function estimating equations. *Am J Kidney Dis.* 2009;54:33–42.)

presence of large numbers of RTE cells or granular or cellular casts.

CRYSTALS

A large variety of crystals seen in the urine may be of diagnostic value in the evaluation of kidney stones and AKD and CKD (Table 23.10). Crystals precipitate in urine as a result of supersaturation and may reflect increased production of the solute or presence of promoters or absence of inhibitors of crystallization (see Chapter 40). Uric acid, calcium oxalate, and calcium phosphate crystals are common without overproduction due to dehydration, extremes of pH, or cooling of the sample and may not be clinically significant. However, uric acid crystals may cause acute uric acid nephropathy as part of tumor lysis syndrome, and calcium oxalate crystals may indicate ethylene glycol

poisoning or hyperoxaluria. Many drugs may cause crystals in the urine that have unusual shapes, which may sometimes be associated with acute tubulointerstitial disease, such as acyclovir or indinavir.¹⁶⁴

MICROORGANISMS

Bacteria are commonly seen because of contamination or infection. Fungi such as *Candida*, protozoa such as *Trichomonas*, and parasites such as *Schistosoma* may also be seen.

APPROACH TO EVALUATION OF HEMATURIA

Hematuria is common and has many causes. Its prevalence varies across different countries. Several large cohort studies have shown that asymptomatic microscopic

Table 23.9 Common casts

Cast	Main clinical association(s)
Hyaline	Normal and in kidney disease
Fine granular	Normal and in kidney disease; casts contain protein aggregates
Coarse granular	Kidney disease of any cause; casts contain degenerated cellular elements
Waxy	Kidney disease of any cause; casts result from degeneration of other casts
Broad	Chronic kidney disease; casts reflect hypertrophy of the tubule
Fatty	Kidney disease with heavy proteinuria
Red blood cell	Proliferative glomerulonephritis; casts result from glomerular bleeding
White blood cell	Inflammatory tubulointerstitial disease
Tubular epithelial cell	Often the result of active acute kidney diseases such as acute tubular necrosis, proliferative glomerulonephritis, acute interstitial nephritis, thrombotic microangiopathy; casts formed from the aggregation of desquamated cells of the tubule

From Fogazzi GB, Verdesca S, Garigali G. Urinalysis: core curriculum 2008. *Am J Kidney Dis.* 2008;51:1052–1067.

hematuria has been associated with the development of CKD including kidney failure; thus it is often considered a marker of kidney damage (Table 23.8).^{165,166} Hematuria in the context of anticoagulation is most likely due to an underlying cause and therefore requires investigation. Warfarin-induced nephropathy, in which obstruction of tubules by RBC casts causes hematuria and AKI, has also been described.^{167,168} Recently, RNA-based vaccinations have been associated with transient gross hematuria. The epidemiology is not known but hypothesized to be due to unmasking an underlying glomerular disease such as IgA nephropathy.¹⁶⁹

The approach to evaluation of hematuria considers causes of red discoloration of urine, symptoms or settings of transient conditions, and consideration of kidney disease or genitourinary malignancy as causes of persistent, asymptomatic, microscopic hematuria (Fig. 23.6). The evaluation begins with urine dipstick and automated microscopy to confirm the finding of RBCs, followed by a history, physical examination, and investigations for such likely causes. Transient hematuria is common and may be due to strenuous exercise, nephrolithiasis, urinary tract infection, anticoagulation, trauma, or contamination of urine with menses. A repeat urinalysis should be taken in 4 to 6 weeks after appropriate treatment to ensure resolution of hematuria. In asymptomatic patients, or in initially symptomatic patients with persistent asymptomatic hematuria after treatment, further evaluation by a nephrologist for presence of kidney disease is indicated. Dysmorphic RBCs, RBC casts, or albuminuria suggest glomerular disease. A decreased eGFR in the absence of these findings suggests a tubulointerstitial, vascular or cystic kidney disease, which

should be investigated by imaging, biopsy, or genetic testing.¹⁷⁰ If kidney disease is not suspected, patients should be risk stratified for the likelihood of genitourinary tract malignancy.¹⁵⁷ The American Urologic Society and Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction recommend risk stratifying patients into low, medium, or high risk on the basis of age, sex, smoking history, and level of hematuria. Evaluation by a urologist and appropriate imaging is recommended for further investigation for patients at medium or high risk. Those at low risk can undergo a repeat urinalysis in 4 to 6 weeks to assess resolution of hematuria before further evaluation.

CLINICAL APPLICATIONS

SIGNIFICANCE

The significance of each of the aspects of laboratory assessment of kidney disease discussed in this chapter reflects its relationship to abnormalities in kidney function and structure, as reflected by decreased GFR and markers of kidney damage, respectively (see Table 23.1).

DEFINITION AND STAGING OF KIDNEY DISEASE

Kidney disease is defined as structural or functional disorders of the kidney with implications for health, with AKD and CKD, respectively defined according to the duration, <3 months and ≥3 months, respectively. AKI is a subset of AKD with onset within 7 days (see Table 23.1). Decreased GFR fulfills the functional criterion: estimated or measured GFR for CKD, changes in serum creatinine or oliguria for AKI, and either criterion for AKD. Markers of kidney damage fulfill the structural criterion for AKD and CKD: Laboratory abnormalities discussed in this chapter include albuminuria, tubular proteinuria, impaired urinary concentration or acidification, and presence of RTE cells, cellular or coarse granular casts, or lipid in the urine sediment. Laboratory abnormalities discussed in other chapters include tubular disorders of electrolyte, mineral, and water metabolism. Markers of kidney damage are not required and are not sufficient as criteria for AKI. The thresholds for disease definition do not vary according to age, clinical characteristics, or comorbidity.

Graded abnormalities in serum creatinine, urine output, GFR, and albuminuria are used to define stages of severity of AKD and CKD. Staging is important as each GFR and albuminuria category is linked to possible treatments in AKI and CKD.¹⁻³ Guidelines recommend adopting a consistent nomenclature for GFR and ACR categories to replace alternative terms.¹⁷¹ Preferred terms for GFR categories are as follows: G1, normal to increased; G2, mildly reduced; G3 (G3A and G3B), moderately reduced; G4, severely reduced; and G5, kidney failure. Preferred terms for albuminuria categories are as follows: A1, normal to mild; A2, moderate; and A3, severe. A1 has been subdivided for epidemiologic studies into <10 mg/g (normal) and 10 to 29 (mild), and A3 has been subdivided into nonnephrotic (300–2000 mg/g) and nephrotic range (>2000 mg/g). (NOTE: The term “microalbuminuria,” referring to dipstick negative, elevated albuminuria, approximately equivalent to A2, and

Table 23.10 Common crystals and their appearance

Crystal	Appearance	pH range	Birefringence
Endogenous			
Ammonium biurate	Yellow-brown spheres with spicules, thorn apples	5.5-7.0	Strong
Bilirubin crystals	Yellow needle-like crystals, attach to cell surfaces	5.5	Weak
Calcium carbonate	Dumbbells, thick rods, 4-leaf clover	7.0	Strong
Calcium oxalate	Monohydrate: colorless ovoid, dumbbells, rods	5.4-6.7	Strong
	Dihydrate: colorless bipyramidal	5.4-6.7	Weak
Calcium phosphate	Prisms, sticks, needles, stars, rosettes in isolation or in aggregates	6.7-7.0	Strong
Cholesterol	Thin plates with well-defined edges	5.5	Negative
Cystine	Colorless hexagonal plates with irregular sides	5.5	Weak
2,8-Dihydroxyadenine	Reddish-brown round with central spicules and dark outline	5.5-7.0	Maltese cross
Leucine	Yellow-brown spheres with concentric striations	5.5-6.5	Maltese cross
Tyrosine	Colorless to yellow thin needles in bundles or rosettes	5.5-6.5	Strong
Triple phosphate (struvite)	Trapezoids, prisms, feather-like, “coffin lids”	6.2-7.0	Strong
Drug-related			
Acyclovir	Thin needles with sharp or blunt ends	5.5-7.0	Strong
Amoxicillin	Colorless thin needles, broom/brushlike	5.5-6.5	Strong
Atazanavir	Thin needles in isolation or as aggregates	6.0-7.0	Strong
Ciprofloxacin	Colorless needles, stars, fans, sheaves	>7.0	Strong
Methotrexate	Yellow-brown	5.4-6.0	Strong
Sulfadiazine	Amber as shocks or sheaves of wheat, shells	5.5	Strong
Triamterene	Brown and other colors (green/orange/red); spheres	5.5	Maltese cross
Vitamin C (calcium oxalate)	Same as for monohydrated calcium oxalate	5.4-6.7	Strong

From Cavanaugh C, Perazella MA. Urine sediment examination in the diagnosis and management of kidney disease: core curriculum 2019. *Am J Kidney Dis*. 2019;73(2):258–272.

“macroalbuminuria” or “clinical proteinuria,” referring to dipstick positive proteinuria, approximately equivalent to A3, are no longer recommended.) Nephrotic-range proteinuria refers to protein excretion of >3.5 g/24 hours. Nephrotic syndrome is defined as urine total protein excretion >3000 or 3500 mg/day, hypoalbuminemia, hyperlipidemia, and edema.

Measures of decreased GFR and increased proteinuria have been extensively evaluated in AKD and CKD. For example, lower eGFR, more rapid decline in eGFR or rise in serum creatinine, more severe oliguria, higher urine ACR or PCR, and greater rise in urine ACR and PCR are each associated with higher risks of KFRT, a broad spectrum of treatable complications including CVD and death.¹⁷²⁻¹⁷⁶ In addition, CKD is a risk factor for AKI, and AKI is a risk factor for development and progression of CKD. These risk relationships provide the rationale for GFR and albuminuria thresholds used for disease definitions, as well as for the GFR and albuminuria categories used for staging of disease severity and guideline-directed care.

Of note, decreased eGFRcys and decreased eGFRcr-cys demonstrate stronger associations with some outcomes than decreased eGFRcr, especially in older people.¹⁷⁵ Even after adjusting for age, compared with a small eGFRdiff (eGFRcys minus eGFRcr), a larger negative eGFRdiff (eGFRcr higher than eGFRcys) is associated with a higher risk of many adverse outcomes of kidney disease, while a larger positive

eGFRdiff (eGFRcr lower than eGFRcys) is associated with a lower risk of adverse outcomes. These findings suggest that the difference in risk of adverse outcomes is likely due to differential confounding of the relationship of GFR to adverse outcomes by factors associated with the non-GFR determinants of the markers (positive confounding by high eGFRcys vs. negative confounding by high eGFRcr).

GFR and albuminuria are used in risk prediction instruments to inform key clinical decisions, improve the patient–health care provider dialogue, and enable personalized care for persons with CKD (see [Chapter 19](#)).^{177,178} GFR and albuminuria can also be incorporated into existing CVD risk prediction instruments to modify CVD risk predictions.^{179,180} More frequent use of these and other instruments holds promise for improving a broad spectrum of clinical outcomes across the full range of GFR and albuminuria.¹⁸¹

SCREENING AND DETECTION OF KIDNEY DISEASE IN POPULATIONS AT INCREASED RISK

AKD and CKD are common, but most patients with kidney disease are not aware of their disease. Early detection could enable initiation of treatments to prevent progression and complications of AKD and CKD. Numerous guidelines suggest that adults at increased risk for CKD should be evaluated using serum creatinine for eGFR and tests for

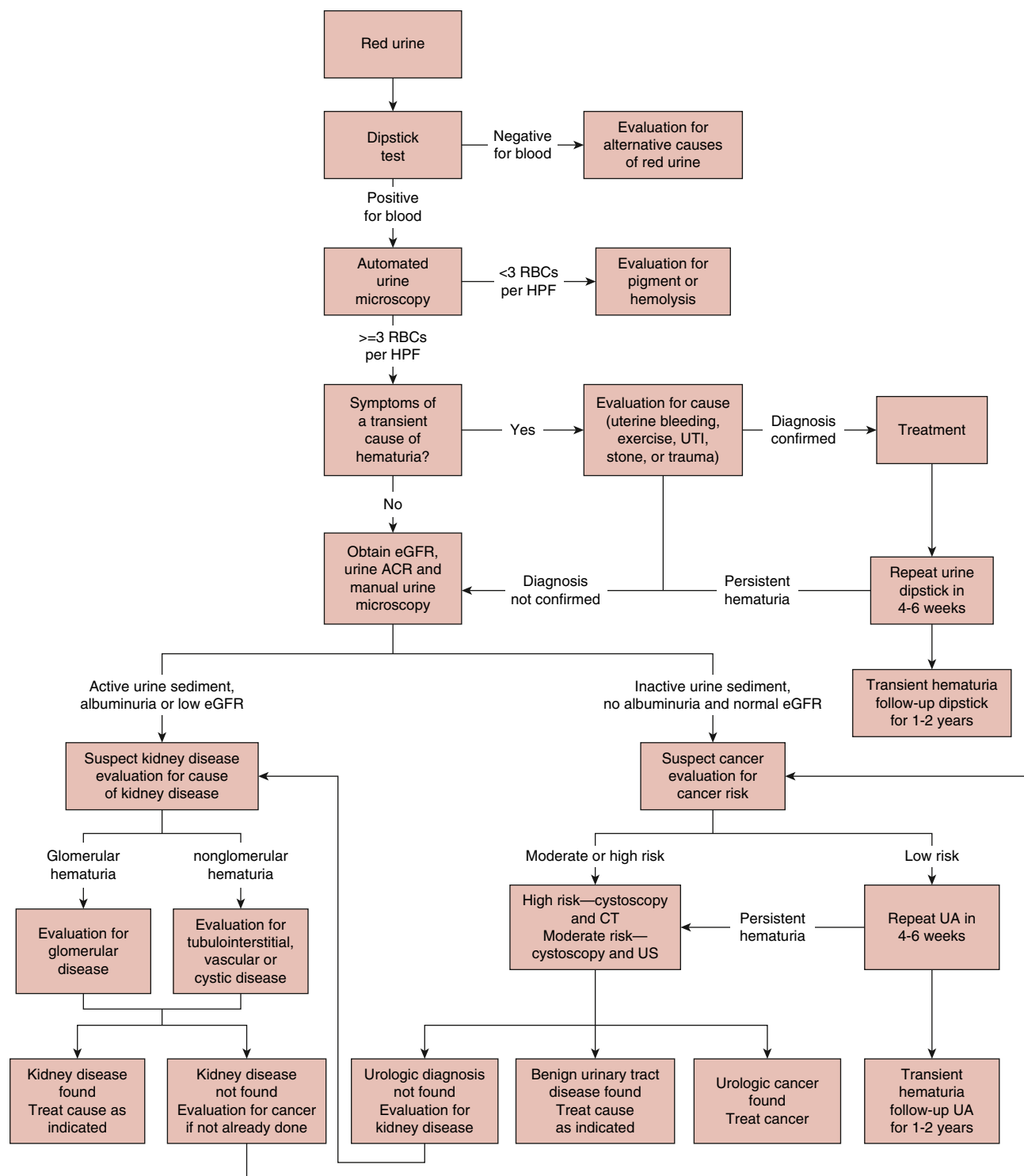


Fig. 23.6 Evaluation of hematuria. The algorithm describes our approach to the evaluation of hematuria. Hematuria should first be confirmed via dipstick and automated urine microscopy, followed by evaluation for transient causes of hematuria (such as uterine bleeding, exercise, urinary tract infection, stone, or trauma). The patient should then receive a stepwise evaluation for kidney disease, followed by risk stratification/evaluation for bladder cancer and urologic causes of hematuria. High-risk patients are ≥ 60 years old and have >30 -pack-year smoking history, >25 red blood cells/HPF on a single urine microscopy, or history of gross hematuria. Moderate-risk patients are women 50–59 years old, men 40–59 years old, people with a 10- to 30-pack-year history of smoking, those with 11 to 25 RBC/HPF on a single urine microscopy, and other relevant history. Negative workups for cancer or urologic disease should prompt a formal workup for kidney disease and vice versa.

urine protein,¹⁸²⁻¹⁸⁵ although the frequency at which testing should occur is either not specified or inconsistent. However, screening of the general population for CKD is not currently

recommended in the United States and several other countries. Nonetheless, there are selected examples where population-based screening is conducted, such as screening

Table 23.11 Use of laboratory assessment in diagnosis of cause of kidney disease

Kidney parenchymal involvement	Presence of “active disease” on pathologic examination (examples below)	GFR	Proteinuria	Dipstick	Possible microscopy findings
None	No	NA	A1	—	—
Glomerular diseases	Yes (proliferation)	NA	A3 (may be nephrotic range)	Heme	Dysmorphic RBC, RBC casts, RTE, granular casts
	No	NA	A3 (may be nephrotic range)		Lipid, RTE, granular casts
Tubulointerstitial disease	Yes (inflammation)	NA	A2/A3 (may have nonalbumin proteinuria)	Leukocyte esterase	WBC and WBC casts, nondysmorphic RBC, RTE, granular casts
	No	NA	A1/A2 (may have nonalbumin proteinuria)	May have defects in urine concentration and acidification	RTE, granular casts
Vascular diseases	Yes (thrombotic microangiopathy)	NA	A2/A3	Heme	Nondysmorphic RBC, RTE, granular casts
Other	No	NA	A1/A2		RTE, granular casts
	No	NA	A1/A2		RTE, granular casts

None includes decreased kidney perfusion or acute obstruction. Nonproliferative glomerular diseases include diabetic nephropathy, podocytopathies, and membranous nephropathy. Inflammatory tubulointerstitial diseases include kidney diseases associated with urinary tract disorders including infection, stones, and chronic obstruction. Other refers to diseases that may affect any location in the kidney parenchyma, such as cystic kidney diseases or congenital anomalies of the kidneys and urinary tracts (CAKUT). Genetic diseases are not considered a separate category because diseases in each category are now recognized as having genetic determinants. Drug toxicity is not considered a separate category but can affect all locations in the kidney parenchyma. Pathologic findings in kidney transplant recipients are not fundamentally different from those in native kidney disease, except for the occurrence of acute and chronic tubulointerstitial and vascular rejection.

NA, Not applicable as level of GFR is generally not helpful in ascertaining cause of disease. Exceptions include minimal change disease, in which GFR is normal, and acute tubular necrosis and tubulointerstitial transplant rejection, in which GFR is reduced. Proteinuria, dipstick, and microscopy categories reflect most likely findings; individual patients may not fit into these categories. A1, A2, and A3 refer to categories for albuminuria.

RBC, Red blood cell; RTE, renal tubular epithelial cell; WBC, white blood cell.

of school-age children in Japan for hematuria by dipstick because of the high prevalence of glomerulonephritis (IgA nephropathy), and screening of pregnant women for proteinuria by dipstick is recommended for early detection of preeclampsia.

CAUSES OF KIDNEY DISEASE

Level of GFR is generally not helpful in determining the cause of AKD or CKD. Level of albuminuria, total proteinuria, dipstick, and microscopy are useful to identify glomerular diseases (albuminuria), tubulointerstitial disease (tubular proteinuria), proliferative glomerulonephritis (dysmorphic RBC and RBC casts), inflammatory tubulointerstitial nephritis (WBC and WBC casts), and thrombotic microangiopathy (nondysmorphic RBC with RTE cells) (Table 23.11). Magnitude of albuminuria (>300 mg/day) and urine albumin-to-urine protein ratio (>0.4) may be useful tools in differentiating

between glomerular and tubulointerstitial diseases. These findings, together with history and physical examination, help direct ordering of other laboratory tests, such as serologic or genetic testing or kidney biopsy (see Chapters 24, 25, and 45).³

DRUG DOSING IN KIDNEY DISEASE

Many medications are eliminated by the kidneys, largely by glomerular filtration but also by tubular secretion. For such medications, dosage adjustment for the level of GFR is necessary to avoid treatment failure or adverse drug events. The CG equation has traditionally been used in past pharmacokinetic studies that serve as the basis for the drug dosing and continues to be widely used despite its known limitations (see earlier). KDIGO recommends using the best available method to evaluate GFR for an individual patient to guide drug dosing, consistent with recognition by major regulatory agencies that “any contemporary,

widely accepted, and clinically applicable estimating GFR equation is considered reasonable to assess kidney function in pharmacokinetic studies.”^{186,187}

ASSESSMENT OF KIDNEY DISEASE PROGRESSION AND REMISSION

Progression of kidney disease refers to worsening of kidney disease severity over time; remission refers to improving over time. Changes in serum creatinine in AKI, changes in eGFR, total urine protein, urine albumin in CKD, and changes in hematuria in glomerular disease have been used to evaluate progression and remission. When evaluating changes over time in a laboratory measure, it is important to consider whether the change reflects preanalytical (e.g., patient preparation, time of day), analytical (laboratory measurement methods), or biological (changes in the true parameter) variability. For changes in eGFR, it is also important to consider whether the change reflects a change in true GFR or a change in the non-GFR determinants of the filtration marker. For change in ACR, it is also necessary to consider whether the change reflects a change in AER or a change in CER.

Regulatory bodies now accept 30% and 40% GFR decline and mean changes in GFR (slopes) over 2 to 3 years in various settings, in addition to the established surrogate endpoint of a doubling of serum creatinine (equivalent to a 57% eGFRcr decline).^{188,189} Changes in albuminuria are also used as endpoints in select circumstances.¹⁹⁰

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QUESTIONS

1. A 59-year-old, 100-kg, 193-cm-tall man has a serum creatinine concentration of 1.5 mg/dL. He works out daily at the gym. Estimated glomerular filtration rate (eGFR_{cr}) using the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) creatinine equation and standardized serum creatinine is 50 mL/min/1.73 m², but measured GFR is 90 mL/min/1.73 m².

What factor most likely explains the underestimation of measured GFR?

- a. Nonsteady state of serum creatinine
- b. Drug-induced inhibition of tubular secretion of creatinine
- c. Decreased extrarenal elimination of creatinine
- d. Increased creatinine generation from large muscle mass or diet

Answer: d

Rationale: Endogenous filtration markers are determined by GFR, as well as physiologic processes other than GFR that affect the plasma concentration of the endogenous filtration marker. These processes include generation, renal tubular reabsorption and secretion, and extrarenal elimination of the marker, and collectively, they are termed non-GFR determinants. For creatinine, the most important of the non-GFR determinants is variation of its generation according to muscle mass and/or injury, physical activity, and consumption of meat or creatinine. This patient has higher muscle mass than someone else of the same age and sex, accounting for an error when compared with the measured GFR.

2. A 60-year-old woman with a 20-year history of diabetes, hypertension, and 30-pack-year smoking history is referred to you for evaluation of possible CKD. She has diabetic retinopathy, a previous myocardial infarction, and a right transtibial amputation. Her body mass index is 27 kg/m², creatinine is 1.38 mg/dL, eGFR_{cr} is 44 mL/min per 1.73 m², and albumin-to-creatinine ratio is 780 mg/g creatinine. Her medications are lisinopril, furosemide, carvedilol, aspirin, atorvastatin, metformin, and multiple injections of short and long-acting insulin. Her doctor is considering adding an SGLT-2 (sodium-glucose cotransporter-2) inhibitor.

What is the best next step in evaluation and management of this patient?

- a. Iohexol plasma clearance
- b. Measure cystatin C to compute eGFR_{cys} and eGFR_{cr-cys}
- c. 24-hour urine for creatinine clearance
- d. Repeat the eGFR using creatinine (eGFR_{cr})
- e. Start an SGLT2 inhibitor
- f. Start a GLP-1 receptor antagonist

Answer: b

Rationale: In this patient, a more accurate value for GFR is needed before treatments that are dosed or indicated based on the level of GFR, such as continuation of the metformin or starting an SGLT-2 inhibitor. The patient has an amputation, which may cause eGFR_{cr} to overestimate her true GFR. Her presumed high fat mass and ongoing smoking raise concerns about the accuracy of cystatin C–based

estimates but eGFR_{cys} is the next best method of estimating GFR given it is relatively easy to obtain. Iohexol plasma clearance could also be used for more accurate estimation of GFR is not the next best step because it would require a specialized center and multiple blood draws.

You draw a cystatin C level, and eGFR_{cys} is 25 mL/min per 1.73 m². Given the discordance between eGFR_{cr} and eGFR_{cys}, eGFR_{cr-cys}, estimated at 32 mL/min per 1.73 m², is the most accurate and could be used for drug-dosing decisions. However, given that these values are on the thresholds for drug-dosing decisions, you could proceed with measured GFR.

3. A 66-year-old man presented to the Kidney Disease Clinic for 6-month follow-up of his known diabetic nephropathy. His baseline eGFR_{cr} is 40 mL/min/1.73 m². The trajectory of eGFR decline has been 2 to 4 mL/min/1.73 m² over the past 4 years. He has had a slowly rising urine albumin-to-creatinine ratio (uACR), and his most recent ACR was 150 mg/g. Today, he reports a 2-month history of fatigue, anorexia, and unintentional weight loss of 5 to 6 kg. Physical examination revealed pulse rate of 80 beats/min and regular blood pressure (140/80 mm Hg), oxygen saturation 95% on room air, clear chest, 1+ pedal edema, and mild pallor.

Laboratory investigations for this clinical visit are as follows: hemoglobin (Hb) 103 g/dL, white cell count 8.2 ($\times 10^3$ /L), platelets 230 ($\times 10^3$ /L), creatinine 4.19 mg/dL, urea 108.1 mg/dL, eGFR_{cr} 20 mL/min/1.73 m², K 5.7 mmol/L, Na 137 mmol/L, HCO₃ (bicarbonate) 20 mEq/L, glucose 1115 mg/dL, and serum albumin 3.2 mg/dL. Urinalysis showed 2+ protein and 3+ red blood cells. Urine microscopy revealed a few red cell casts.

Which of these is the most appropriate response regarding this man's condition?

- a. This is most likely a laboratory error; repeat blood work should be organized.
- b. He has lost a significant amount of weight in the recent past, which has resulted in a rise in creatinine.
- c. He has evidence of significant acute kidney disease and needs further investigation.
- d. This is the natural progression of his underlying diabetic nephropathy.
- e. All of the above.

Answer: c

Rationale: This patient's history and clinical features suggest acute kidney disease due to disease other than diabetic nephropathy. His subacute onset of symptoms with the recent rise in creatinine, active urine sediment, and presence of red cell casts on urine microscopy strongly suggest the diagnosis of a rapidly progressive glomerulonephritis, and he warrants further investigation and management of this condition. Although diabetic nephropathy can rarely progress rapidly, it would be unusual to develop red cell casts in the urine. The weight loss and catabolic state should lead to a decrease in serum creatinine rather than an increase. This result is unlikely to be a laboratory error, especially in the setting of his symptoms, and further investigation and management should not be delayed.

4. A 73-year-old woman was referred to the Kidney Disease Clinic for evaluation of worsening GFR and anemia. Review of her laboratory tests showed hemoglobin (Hb) 8.0 mg/dL (11.0 mg/dL 1 year ago), creatinine 2.04 mg/dL, eGFR-cr 25 mL/min/1.73 m² (67 mL/min/1.73 m² 1 year ago), SUN 42 mg/dL, K 4.6 mEq/L, Na 132 mEq/L calcium 10.8 mg/dL. Her serum protein electrophoresis showed a monoclonal spike in the gamma region. A diagnosis of multiple myeloma is suspected. An office urine dipstick shows 1+ protein, no red cells, no leukocytes, or glucose.

Which of the following statements is true regarding the urine dipstick result?

- The urine dipstick reliably detects albumin but not other proteins such as monoclonal proteins which cause cast nephropathy.
- The urine dipstick is a sensitive marker for tubular and overflow proteinuria.
- A positive result from a semiquantitative test, such as a urinary dipstick test, should prompt further evaluation with a quantitative test.
- Both a and b.
- Both a and c.

Answer: e

Rationale: This patient may have multiple myeloma as evidenced by her history and laboratory findings. Referral to hematology and possible bone marrow biopsy are indicated to confirm the diagnosis. Her GFR is impaired, suggesting kidney involvement. The urine dipstick shows minimal proteinuria, but this test only reliably detects albumin and not other proteins; hence although it can detect glomerular proteinuria, it is a poor test to assess for tubular and overflow proteinuria. A quantitative test including an albumin-to-creatinine ratio (uACR) and a protein-to-creatinine ratio (uPCR) should be requested. In the case of cast nephropathy, one would expect a significant discrepancy in these values with a higher uPCR as compared with the uACR,

suggesting that there is excretion of a protein other than albumin (in this case, light chains).

5. A 58-year-old woman is noted to have dipstick hematuria as part of her insurance physical. No prior urinalysis is available. The patient did not experience any symptoms after thorough questioning. She is a former smoker with 20-pack-year history but quit 15 years ago. You are her primary care doctor and see her 2 months after the initial test and note the presence of hematuria.

What is the best next step?

- Repeat the test.
- Consider her at moderate risk for bladder cancer and obtain an ultrasound.
- Refer her to a nephrologist.
- Obtain GFR and ACR, and ask the laboratory to report on the urine sediment.

Answer: a

Rationale: Transient hematuria is common and may be due to strenuous exercise, nephrolithiasis, urinary tract infection, anticoagulation, trauma, or contamination of urine with menses. A repeat urinalysis should be taken in 4–6 weeks to ascertain if persistent. If your patient does have persistent hematuria, the next step is to evaluate for kidney disease. You do not find any evidence of kidney disease with GFR of 100 and ACR of 15 mg/g. If kidney disease is not suspected, patients should be risk-stratified for the likelihood of genitourinary tract malignancy. The American Urologic Society and Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction recommend risk stratifying patients into low, medium or high risk based on age, sex, smoking history and level of hematuria. She is considered to be medium risk and therefore evaluation by a urologist and appropriate imaging is recommended for further investigation for her.